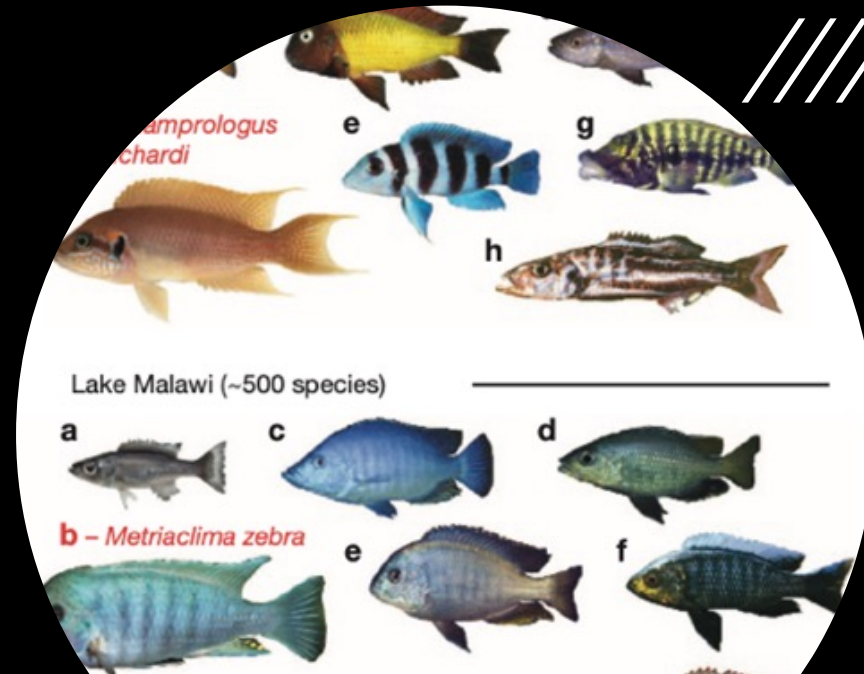
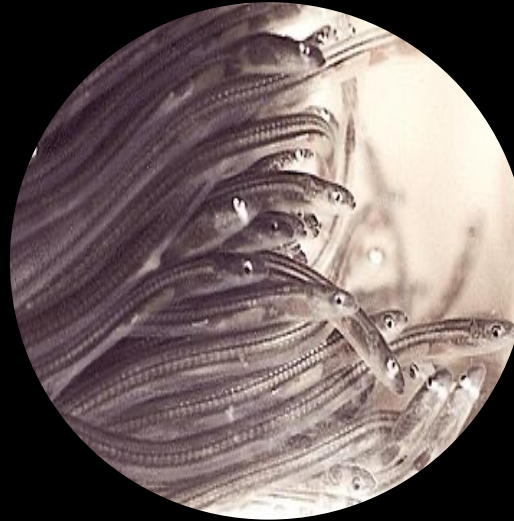


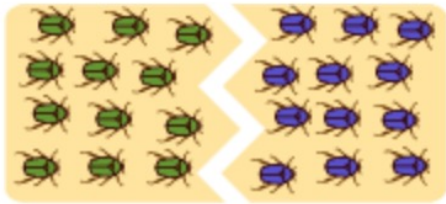
Molecular Methods in Ecology and Evolution 2026



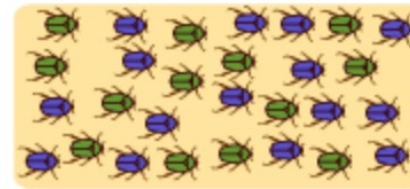
The theme:

Understand the evolutionary and ecological drivers of population divergence and speciation

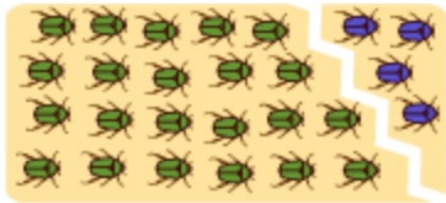
Geographic modes of speciation



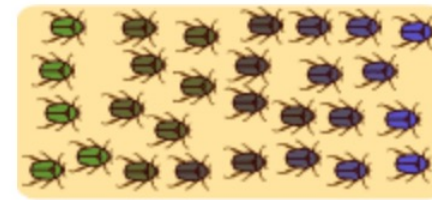
Allopatric: (allo = other) New species formed from geographically isolated populations



Sympatric: (sym = same) New species formed from within the range of the ancestral population



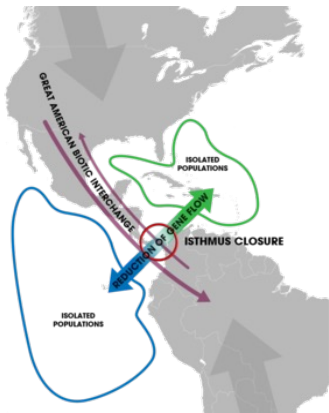
Peripatric: (peri = near) New species formed from a small isolated population at the edge of a larger population



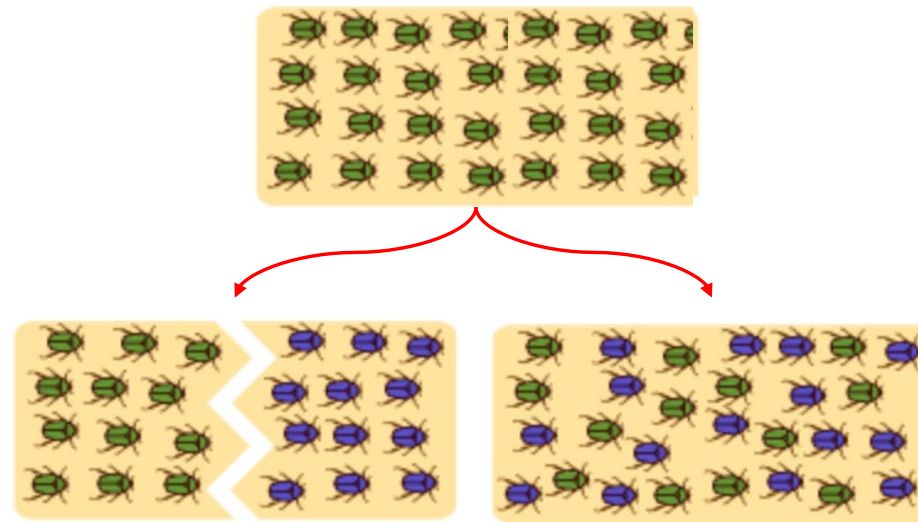
Parapatric: (para = beside) New species formed from a continuously distributed population

The theme:

Understand the evolutionary and ecological drivers of population divergence and speciation



Isthmus of Panama



Geographical / Allopatric
divergence

Ecological / sympatric
divergence



Scincid lizards

The theme:

Understand the evolutionary and ecological drivers of population divergence and speciation

A special case of **parapatric speciation**: Greenish warbler (*Phylloscopus trochiloides*)

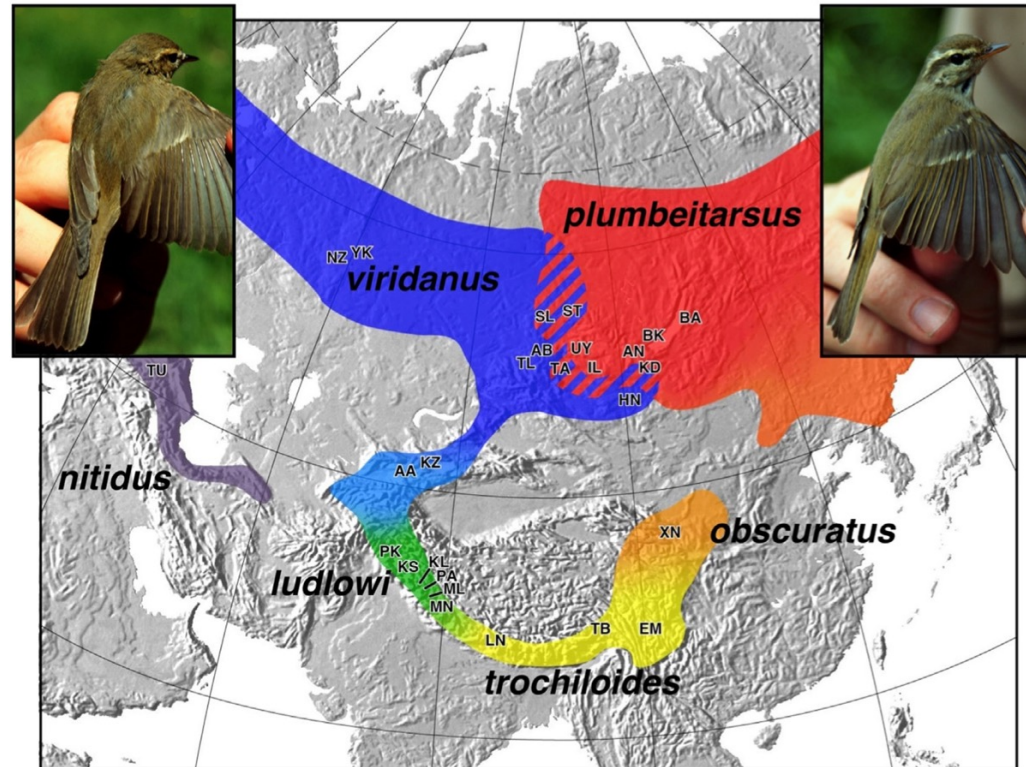


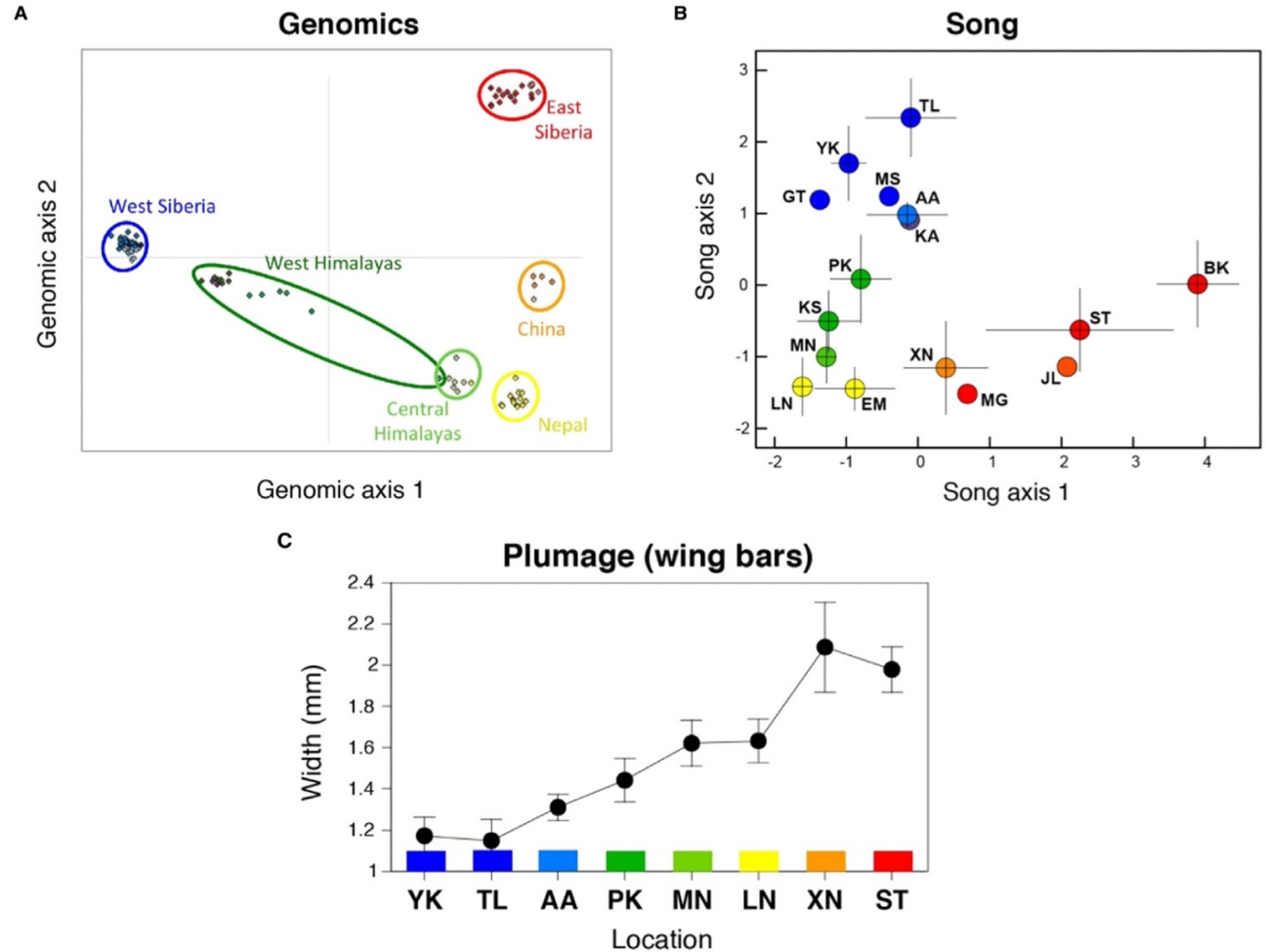
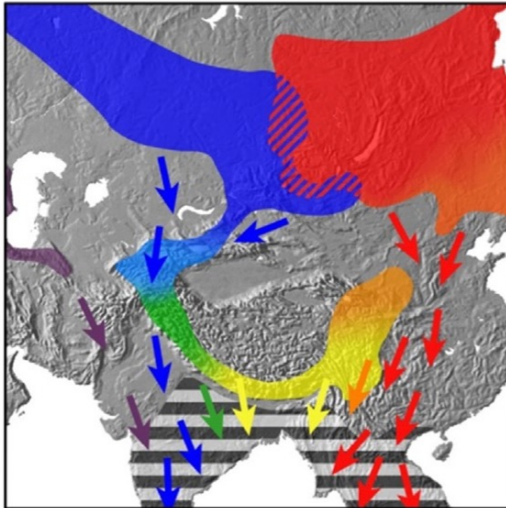
Fig. 4 The breeding range of greenish warblers (*Phylloscopus trochiloides*) in Asia. Subspecies designations according to [Ticehurst \(1938\)](#) are shown with different colors: *viridanus* in blue, *ludlowi* in green, *trochiloides* in yellow, *obscuratus* in orange, *plumbeitarsus* in red, and *nitidus* (outside of the main ring) in purple. Photos show the difference in wing bars between *viridanus* (upper left, with a single wing bar), and *plumbeitarsus* (upper right, with two wing bars).

The theme:

Understand the evolutionary and ecological drivers of population divergence and speciation

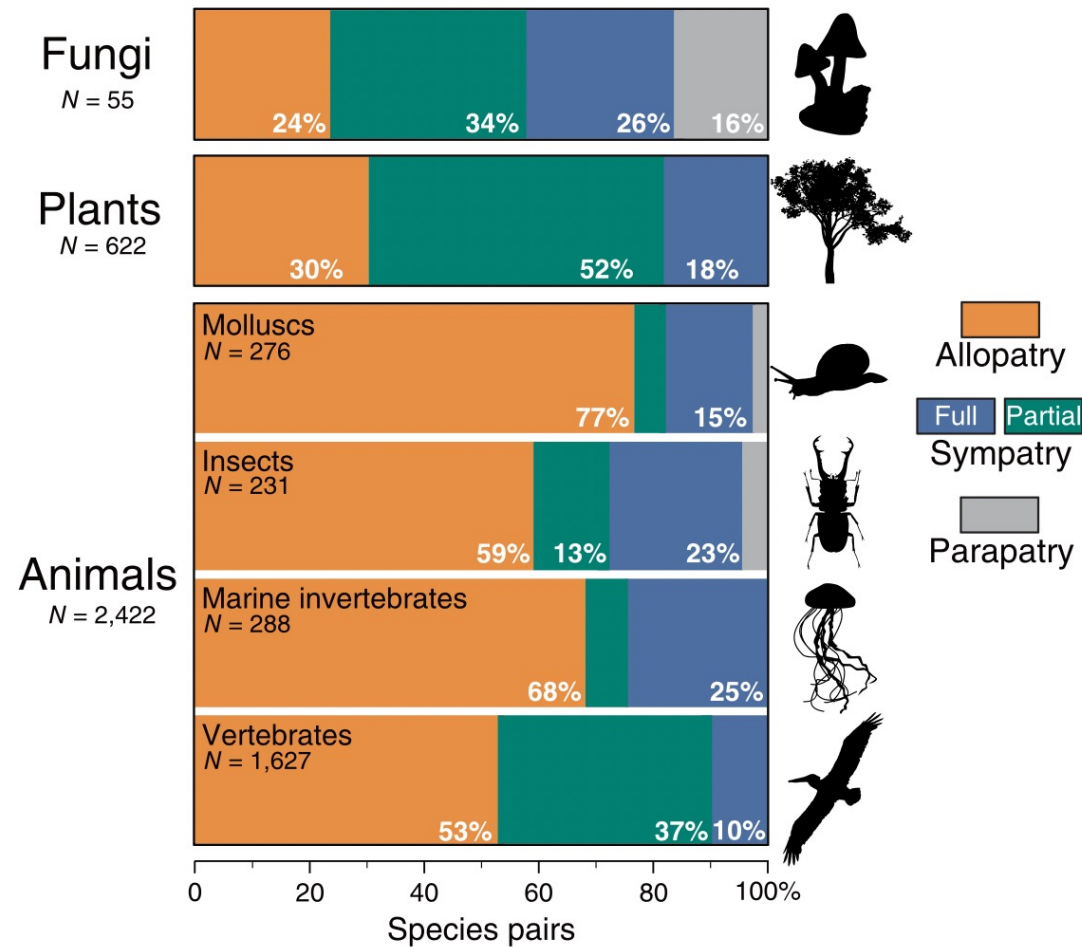
Patterns of genomic and phenotypic variation show strong differentiation between the mayor two forms, but gradual / stepwise variation through the chain of populations to the south.

Migratory routes



The theme:

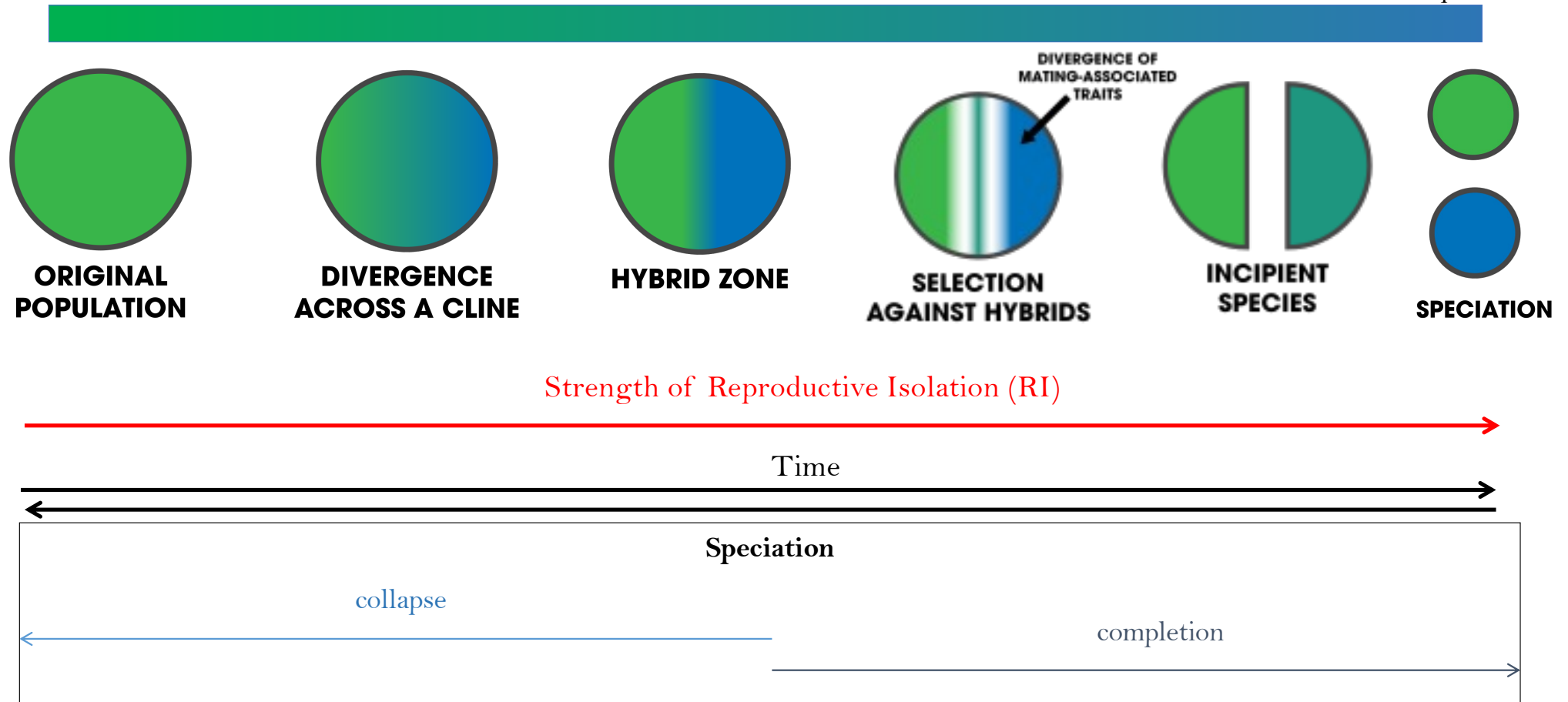
Understand the evolutionary and ecological drivers of population divergence and speciation



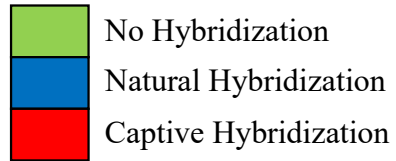
The Speciation Continuum

Panmictic
population

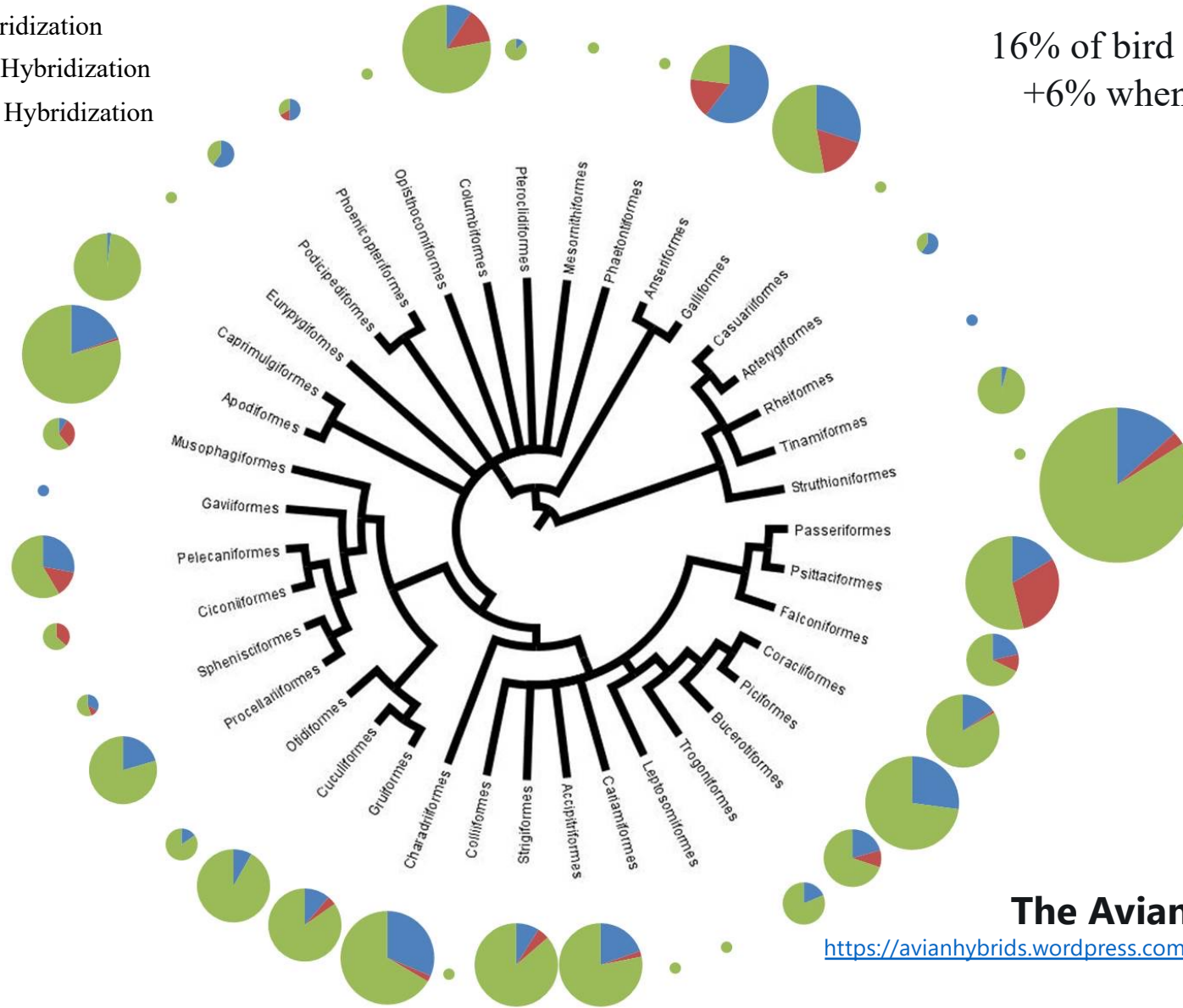
Two isolated
species



The speciation process is dynamic, bidirectional and continuous.



16% of bird species hybridize in the wild
+6% when including captive hybrids



The Avian Hybrids Project

<https://avianhybrids.wordpress.com/2017/12/01/hybridization-in-birds-the-trilogy/>

Ottenburghs et al. (2015) *Ibis*

The goal:

To give you an insight to how biologists use specific techniques and tools to tackle ecological and evolutionary questions

To gain experience on some of these approaches, and the application of the scientific method to real-world systems.



- Marker gene amplification and sequencing
- Restriction site Associated DNA sequencing (RADseq)



- Phylogenetic reconstruction
- Population genetics
- Population structure analyses
- Genomic-scale analyses of natural selection
-

Project 1.

A mysterious frog



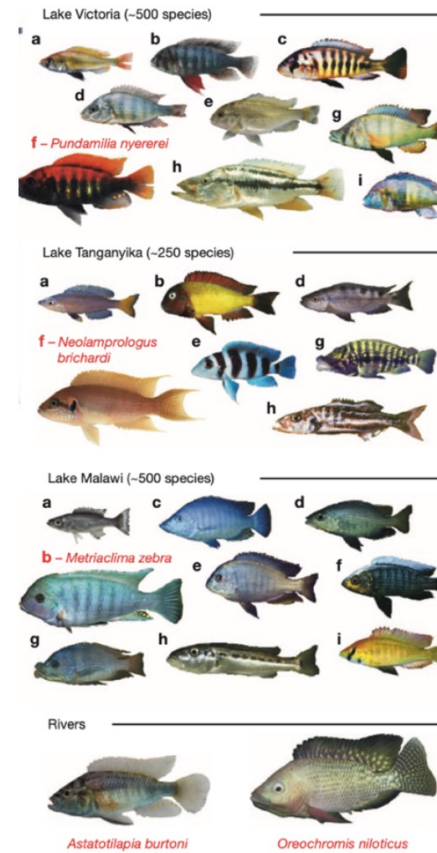
Project 2.

Hylid frogs



Project 3.

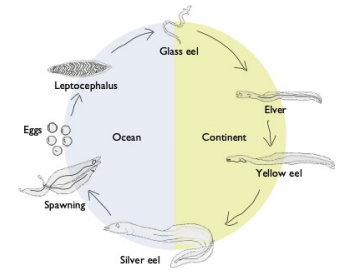
Cichlids



Brawand et al., 2014

Project 4.

European eel



Schedule

Time	Monday Sept 28	Tuesday Sept 29	Wednesday Sept 30	Thursday Oct 1st	Friday Oct 2nd
08:00 - 08:45	DNA extraction	PCR Gel preparation	Sequencing prep	Lecture	
09:00 - 09:45					
10:15 - 11:00	Lecture	Lecture		Lecture	
11:15 - 12:00					
12:15 - 13:00					
13:15 - 14:00	TP Introduction	Electrophoresis	Lecture	Lecture	Lecture
14:15 - 15:00	DNA extraction Quantification Dilution				
15:15 - 16:00					
16:15 - 17:00		PCR purification			

Project 1. →

Time	Monday Oct 5th	Tuesday Oct 6th	Wednesday Oct 7th	Thursday Oct 8th	Friday Oct 9th
08:00 - 08:45	Project 1	Project 2 Cryptic speciation in hylid frogs.	Project 3.	Project 4. The curious case of the European Eel	
09:00 - 09:45	Phylogeny- Sanger seq		Genomic analyses of divergence between Lake Malawi cichlids		
10:15 - 11:00	Lecture			Final questions	
11:15 - 12:00					
12:15 - 13:00					
13:15 - 14:00	Project 1	Project 3. Genomic analyses of divergence between Lake Malawi cichlids	Project 4. The curious case of the European Eel	Personal Research Work	
14:15 - 15:00	Phylogeny- Sanger seq				
15:15 - 16:00	Project 2 : Cryptic speciation in hylid frogs.				
16:15 - 17:00					

↑
Project 2.

↑
Project 3.

↑
Project 4.

Wet-Lab Experiment	POL 203 and 205
Lecture	POL 334
Computer Analyses	POL 334

The report

General introduction
Wet lab – Computer Project 1 - Intro - methods - results & Discuss
Computer lab: Project 2. - Intro - methods - results - discussion - conclusion - references
Project 3. - Intro - methods - results - discussion - conclusion - references
Project 4. - Intro - methods - results - discussion - conclusion - references
General Discussion

max **5 pages** per project
(including text and figures)

max **25 pages** in total
(including text, figures and references)

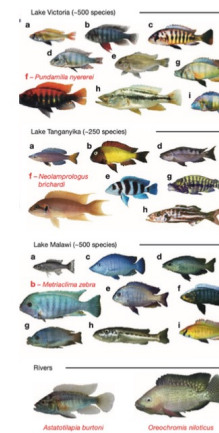
DEADLINE: 27 / 10 / 26

Report grading

See “**Advice on preparing your report**” section in your manual, these guidelines are there to help you write your report, but they will also be the basis of the marking scheme used to grade your reports.

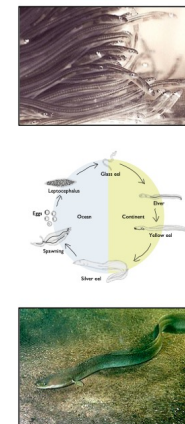
Using molecular approaches to understand the drivers of population divergence and speciation.

Cichlids

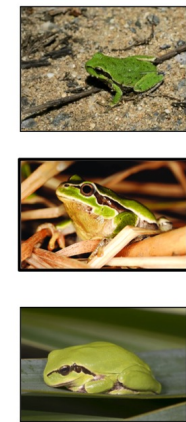


Brawand et al., 2014

European eel



Hylid frogs



Prof. Ian R. Sanders, Dr. Luca Fumagalli, Prof. Nicolas Salamin

Teaching assistants: Dr. Angélica Pulido, Dr. Giulia Maiello, Marion Nyamari, Hugo Corval, Emelyne Gaudichau, Diego Clavijo.

The report

<i>Project 1. Phylogenetic identification of unknown froglet using Sanger sequencing</i>	
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Using Sanger sequencing on the samples you processed in the wet-lab.....	20
Sanger sequence analysis	20
Sequences Alignment	20
Simple phylogeny in R	21
<i>Project 2. Reconstructing the phylogenetic history of Hylid frogs using RAD sequencing</i>	22
.....	
Phylogenetics of the genus <i>Hyla</i> using RAD-seq data.	22
Overview.....	22
Summary of raw RADseq data processing	23
Population genetics of Hylid frogs in R	25
Population structure analyses of frogs in R.....	26
<i>Project 3. Genomic analyses of divergence between Lake Malawi cichlids</i>	28
Population genetics of lake Malawi cichlids.....	29
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DEADLINE: 27 / 10 / 26

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Computer lab: Project 2. - Intro - methods - results - discussion - conclusion - references
Project 3. - Intro - methods - results - discussion - conclusion - references
Project 4. - Intro - methods - results - discussion - conclusion - references
General Discussion

The report

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PCR purification before cycle sequencing reaction.....	14
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Sequences Alignment.....	20
Simple phylogeny in R	21

DEADLINE: 27 / 10 / 26

General introduction

Wet lab – Computer Project 1

- Intro
- methods
- results & Discuss

Computer lab: Project 2.

- Intro
- methods
- results
- discussion
- conclusion
- references

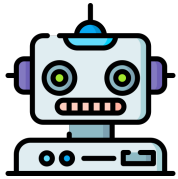
Project 3.

- Intro
- methods
- results
- discussion
- conclusion
- references

Project 4.

- Intro
- methods
- results
- discussion
- conclusion
- references

General Discussion



Use of AI:

The goal:

You **learn** to analyze genomic data and interpreted results to answer biological questions.

DO NOT USE AI

- to write your report for you!
- to generate the interpretation or discussion of results.

YOU CAN USE AI

Clarify concepts that you don't understand

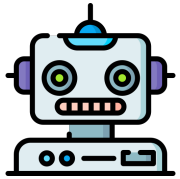
Organize workflow

Generate/**Interpret** bioinformatic code: Use it as a learning tool! – Copilot / Claude

Help in your code if you're stuck

Literature search

Use of AI:



Be critical!
Use it as a leaning tool!

AI tools are often incorrect!!

- Citation for specific claims
- Generating sentences that can not be verified by published studies
- Summarizing a scientific topic wrongly
- over-inflating importance of aspects
- constructing non-logical scientific arguments

Artificial intelligence at UNIL

Website: AI Insight

FAQ: Using AI at UNIL

- [FAQ: AI in Education](#)
- [FAQ: AI for my Studies](#)
- [FAQ: AI in Research](#)

AI_GO: Which AI tool for your data?

AI Training

AI Strategic Unit

AI Expert Group

FAQ: Using AI at UNIL

This FAQ is regularly updated (17/09/2026) to reflect the fast pace of developments in artificial intelligence. For any questions, contact the AI Strategy Unit: iaunil@unil.ch

[Table of contents](#) [Data protection and confidentiality](#)

Can AI be used at UNIL?

What are the integrity values and recommendations for using generative AI at our university?

ChatGPT, Large Language Models and AI: what are we talking about?

What are the main language models currently available?

Which AI tools does UNIL make available to its community?

How can AI be used at UNIL?

What are the risks of using AI tools and what precautions should be taken?

What future directions can we anticipate in the development of this type of tool?

<https://www.unil.ch/unil/en/home/menueinst/universite/intelligence-artificielle/faq.html>

https://github.com/Angelica-Pulido/MMEE-2026/

Angelica-Pulido / MMEE-2026

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Angelica-Pulido

update

52318cb · 2 hours ago

100 Commits

1.Frogs_Sanger	update readme	5 hours ago
2.Frogs_RADseq	update readme	5 hours ago
3.Cichlids	Update 3.0.README_Cichlids.md	11 months ago
4.Eels	Update and rename 3.0.README_Eels.md to 4.0.README...	last year
.DS_Store	update readme	5 hours ago
MMEE_2026_Manual.pdf	update	2 hours ago
README.md	update readme	5 hours ago
Schedule.png	updating data	last week

README

Molecular Methods in Ecology and Evolution - 2026 - University of Lausanne

This is the repository for the master course "Molecular Methods in Ecology and Evolution - 2026 UNIL"
Here you will find all the information and data you will need for the computer analyses of the course.

About

This is the repository for the course "Molecular Methods in Ecology and Evolution - 2025 - UNIL Master"

Readme

Activity

0 stars

2 watching

1 fork

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 2

Angelica-Pulido

Angélica Pulido

sciencesj

AMF.SJ

<https://github.com/Angelica-Pulido/MMEE-2026/>

 **Files**

 main  

Go to file  

- >  1.Frogs_Sanger
- >  2.Frogs_RADseq
- >  3.Cichlids
- >  4.Eels
 -  .DS_Store
 -  MMEE_2026_Manual.pdf
 -  README.md
 -  Schedule.png

MMEE-2026 / README.md 

Preview Code Blame

50 lines (26 loc) · 2.01 KB

  Raw    

- [3.Cichlids](#) - This directory contains the data for the third project.
- [4.Eels](#) - This directory contains the data for the forth project.

Important information

To be able to install and use all packages required you may need to use R version 3.6.x instead of R version 4.x.x. Consider swithcing versions if you have trouble installing the packages. If you already have an R version installed on your computer and want to change it, you can find instructions on how to do it [here](#).

Packages installation

To install all packages required, please run the following commands:

In case you don't have administrator's access to your computer, you can specify where the packages should be installed with the `lib` option in `install.packages()`

```
install.packages("ape", dependencies = TRUE)

install.packages("phangorn", dependencies = TRUE)

install.packages("seqinr", dependencies = TRUE)

install.packages("adeget", dependencies = TRUE)

install.packages("pegas", dependencies = TRUE)

install.packages("hierfstat", dependencies = TRUE)

install.packages("raster", dependencies = TRUE)

if (!requireNamespace("BiocManager", quietly = TRUE))

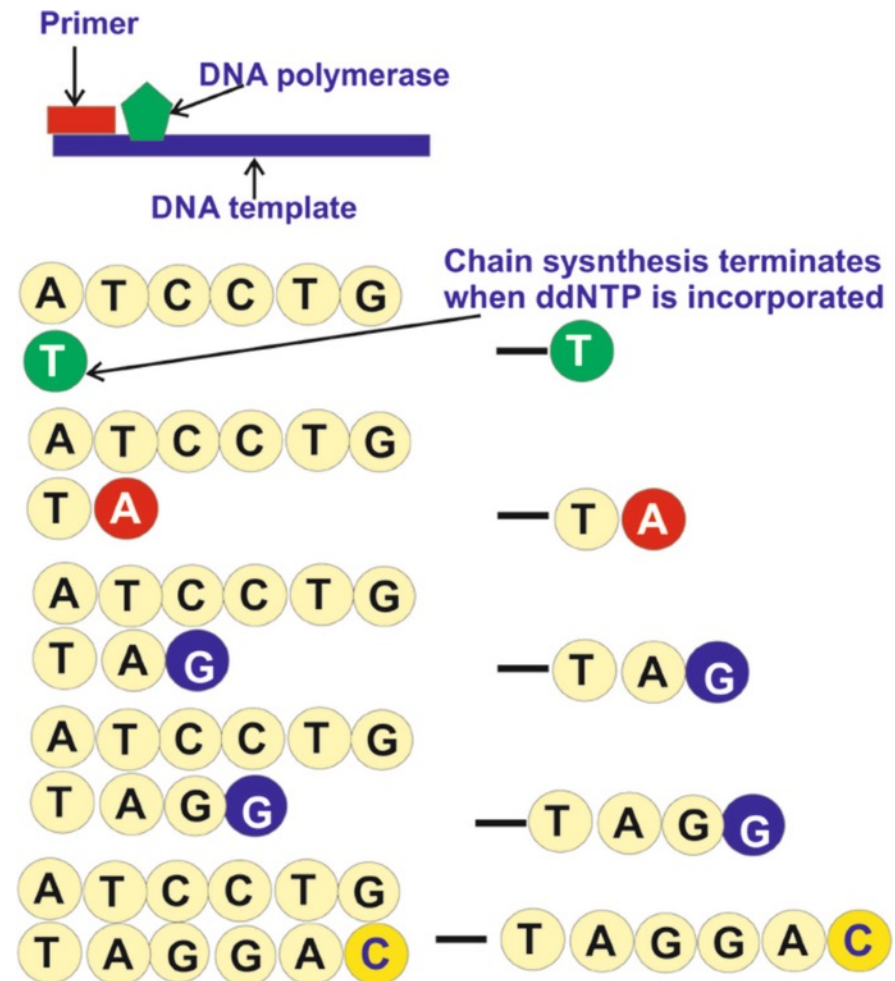
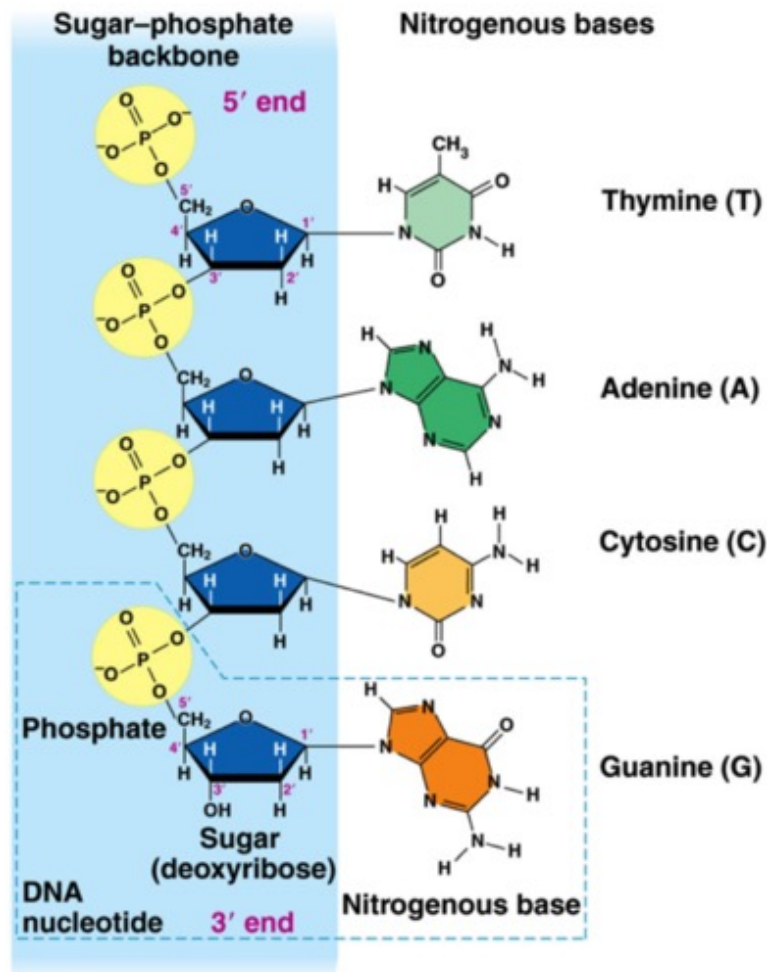
install.packages("BiocManager")

BiocManager::install("LEA", dependencies = TRUE)

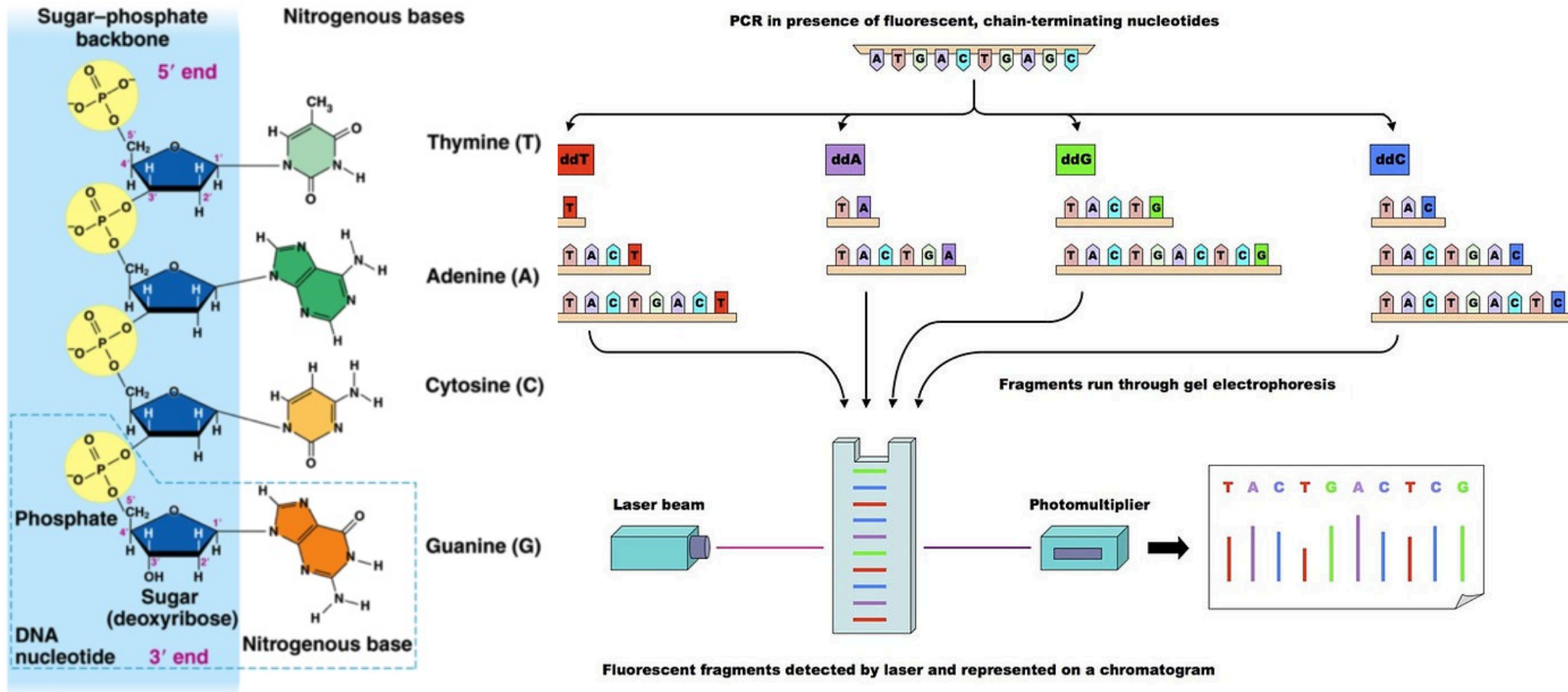
install.packages("outliers", dependencies = TRUE)
```

From Sanger Sequencing to Phylogenetic analysis

What is the principle of Sanger sequencing?

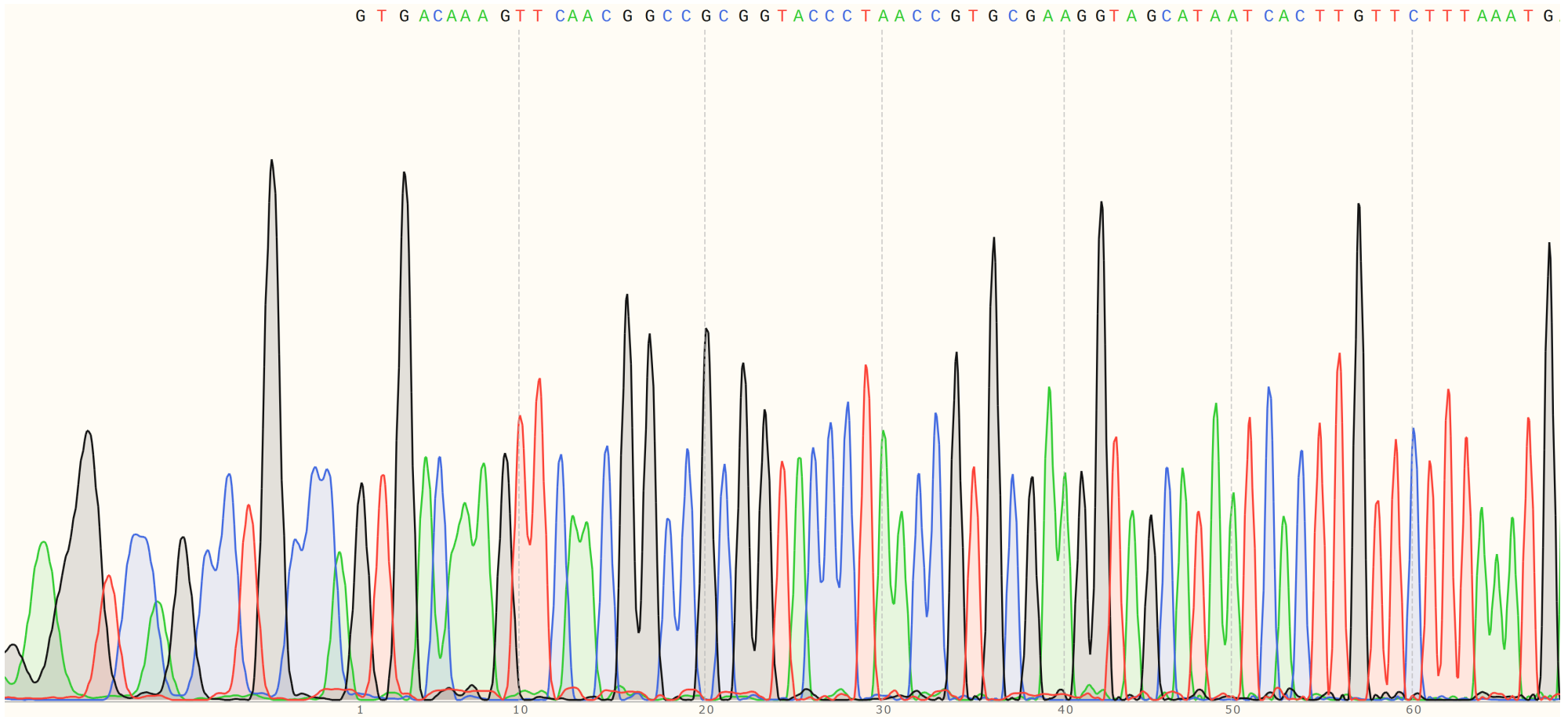


What is the principle of Sanger sequencing?



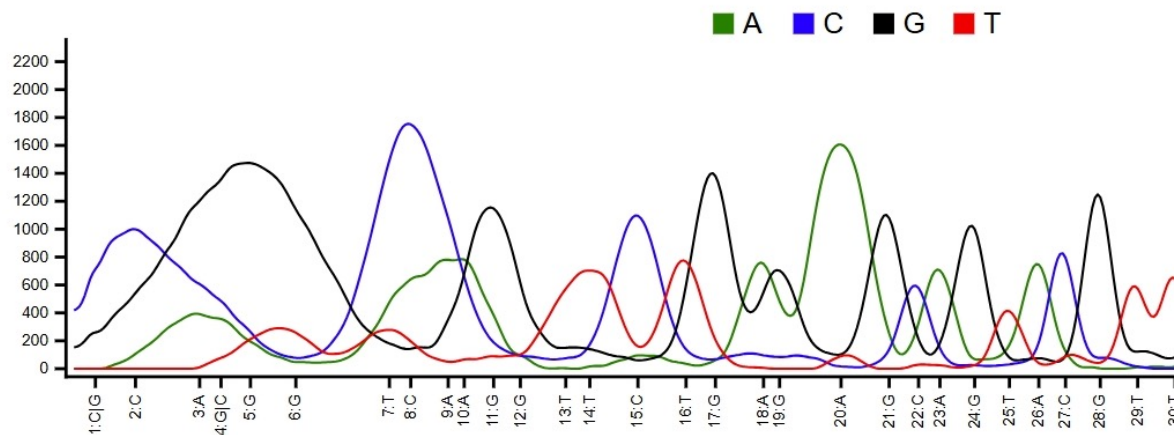
What does the result look like?

Translating chromatogram to “Sequence”



Why two sequences (Forward and Reverse)?

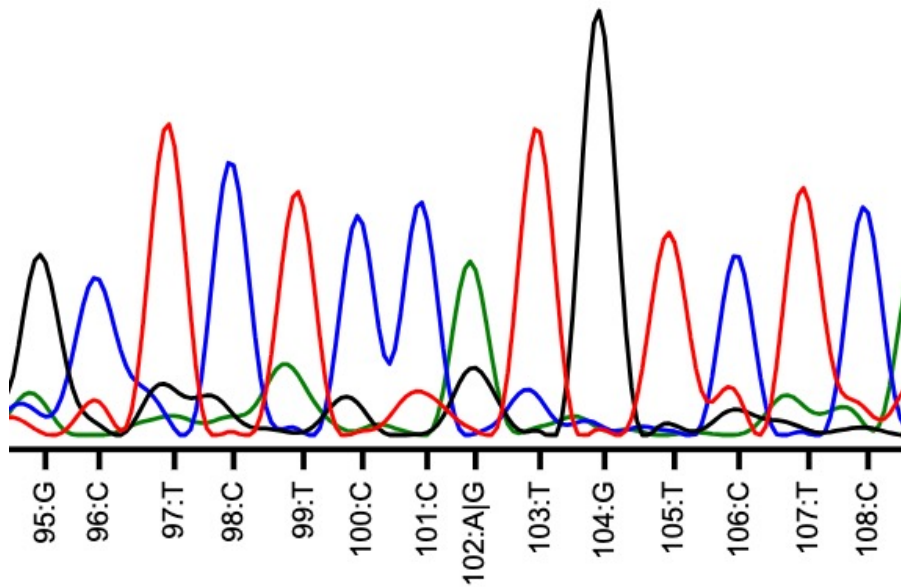
- 1. Quality confirmation:** if the two reads (after reverse-complementing one of them) overlap and match, you have independent confirmation that the sequence is correct
- 2. Full coverage:** Sanger sequencing quality tends to degrade toward the end of a read; having reads from both directions lets you cover the entire fragment at high quality



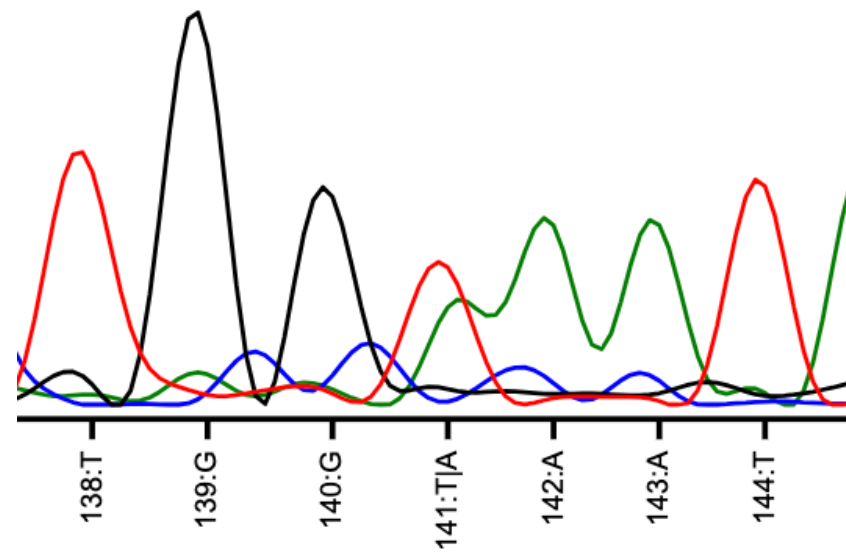
- 3. Resolving ambiguities:** if one read has a low-quality region or an ambiguous base call, the read from the opposite direction can often resolve that position

Ambiguities

■ A ■ C ■ G ■ T



■ A ■ C ■ G ■ T

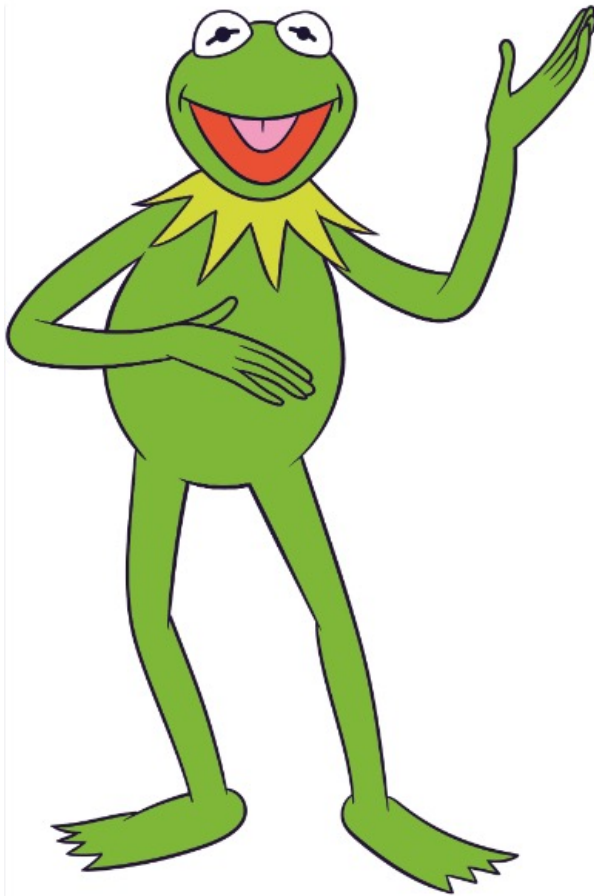


Checking the forward and reverse reads

>Rag-F-LL

AATCTTTGAAGAGT**W**GCTTGCTGATGAATCAGACCACG**R**AAACCARACTCTG**Y**CTGATGAATCAGAC

IUPAC nucleotide code	Base
A	Adenine
C	Cytosine
G	Guanine
T (or U)	Thymine (or Uracil)
R	A or G
Y	C or T
S	G or C
W	A or T
K	G or T
M	A or C
B	C or G or T
D	A or G or T
H	A or C or T
V	A or C or G
N	any base
. or -	gap



Project 1

Identifying a mysterious frog species

Using Sanger Sequencing

Check your sequencing results

- Download your sequences: <https://github.com/Angelica-Pulido/MMEE-2026/>
- Check chromatogram (ab1 files): <https://www.gear-genomics.com/teal/>
- Get the complementary sequence of **your reverse read**
- Align the converted reverse sequence with the forward sequence

Computer-lab: Data analyses

Downloading the data for this practical

The first thing you must do is to **download all the data for this course** onto your computer. To do that go here: <https://github.com/Angelica-Pulido/MMEE-2026/> and click on **Clone or Download >> Download ZIP**. Save the folder to your desktop and un-compress it if needed. You should then have all the course materials in the directory:

“C:/Users/yourusername/Desktop/MMEE_2026/” for windows.

“/Users/yourusername/Desktop/MMEE_2026/” for mac or linux.

You can use this path with the `setwd()` function when working in R so you can easily find all of the files you need.

<https://github.com/Angelica-Pulido/MMEE-2026/>

Angelica-Pulido / MMEE-2026

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.DS_Store

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2.Frogs_RADseq

3.Cichlids

4.Eels

.DS_Store

MMEE_2026_Manual.pdf

README.md

Schedule.png

MMEE-2026 / 1.Frogs_Sanger / 1.0.README_Frogs_Sanger.md

Angelica-Pulido Update 1.0.README_Frogs_Sanger.md 03289f8 · last year History

Preview

Code

Blame

12 lines (10 loc) · 710 Bytes

Raw

Project: 1.Frogs_Sanger/

The folders and files for this project are as follow:

Data/Phylo:

A directory with the forward and reverse reads from the sequencing marker for YOUR samples.

- rag1_Sanger_reference.fasta & rhod_Sanger_reference.fasta:
in case your file is empty, you can use these files.

- B2F_rag.ab1 , B2R_rag.ab1 , D2F_rho.ab1 , D2R_rho.ab1
forward and reverse reads from the markers used in the weblab, use this files on https://www

- Rag1_Reference_sequences.fasta & Rhod_Reference_sequences.fasta
fasta formatted sequences for many frog species that you can use to built a phylogeny. Use t

https://github.com/Angelica-Pulido/MMEE-2026/

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B2F_rag.ab1

B2R_rag.ab1

D2F_rho.ab1

D2R_rho.ab1

Rag1_Reference_sequences.fasta

Rhod_Reference_sequences.fasta

Rhod_Reference_sequences_all_position_...

rag1_Sanger_reference.fasta

rhod_Sanger_reference.fasta

.DS_Store

MMEE-2026 / 1.Frogs_Sanger / Data / Phylo

Angelica-Pulido

update readme

16fc581 · 6

Name	Last commit message
..	
Sanger	updating data
.DS_Store	update readme
B2F_rag.ab1	Add files via upload
B2R_rag.ab1	Add files via upload
D2F_rho.ab1	Add files via upload
D2R_rho.ab1	Add files via upload
Rag1_Reference_sequences.fasta	Add files via upload
Rhod_Reference_sequences.fasta	Add files via upload
Rhod_Reference_sequences_all_position_ali...	Add files via upload
rag1_Sanger_reference.fasta	Add files via upload
rhod_Sanger_reference.fasta	Add files via upload

Checking the forward and reverse reads

Code

Blame

6 lines (4 loc) · 1.06 KB

```
1 >rag1f
2 CAGTACGTTTCTCATTACAATCATGACCATCAGCATTCCAACAACAGTGGGTCTGTTAGAATCTTTGAAGAGTCCAAACCAAACCTCTGAAC
3
4 >rag1r
5 AACTCCTTTCACCTCTGTCACGAAGCTCGTCAGCAGATTCAATTGTGAGGGTTGGCTCTCCATGTCTCATAGCGCTGTAAGTTCTCAGAATGACT
6
```

Checking the forward and reverse reads

Reverse Complement

Reverse Complement converts a DNA sequence into its reverse, complement, or reverse-complement counterpart. You may want to work with the reverse-complement of a sequence if it contains an ORF on the reverse strand.

Paste the raw or FASTA sequence into the text area below.

```
>rag1r
AACTCCTTTCACTCTGTACGAAGCTCGTCAGCAGATTCACTGTGAGGGTTGGCTCTCCA
TGTCTCATAGCGCTGTAAGTTCTCAGAATGACTTCTGGTAATAGAATGGAAGACTAAGTT
CTGAGAAGCTTCCAGCCGAGTAGCATCACAGAGGGTACAAATATAGATAGAGCCTGAGGC
TTCCAGTCCTTCTACTTCTCTAACAAGCTTCTCATCATACCCTGTGCCACGAAAGACAAA
TTTGAAATTTCTGAGGATTCCACCAATTTCCAACATTAGTTCACTGGATTTCATGGCTTC
```

- Convert the DNA sequence into its reverse-complement counterpart.

[\[home\]](#)

The Sequence Manipulation Suite: Reverse Complement
Results for 522 residue sequence "rag1r" starting "AACTCCTTTC".

```
TGTGAGAAGCATGGCAGTGGTCCACCAGTTCCTGAGAAAGCAGTACGTTTCTCATTCA
ATCATGACCATCAGCATTCCCAACAACAGTGGGTCTGTTAGAATCTTTGAAGAGTCCAAA
CCAAACTCTGAACCTTTGCTGCAAACCCCTTTGTCTAATGCTTGCTGATGAATCAGACCAC
GAAACTCTAACAGCCATCTTGAGCCCTTTAATAGCTGAGAGAGAAGCCATGAAATCCAGT
GAACTAATGTTGGAAATTGGTGGAAATCCTCAGAAATTTCAAATTTGTCTTTCGTGGCACA
GGGTATGATGAGAAGCTTGTTAGAGAAGTAGAAGGACTGGAAGCCTCAGGCTCTATCTAT
```



Multiple Sequence Alignment by CLUSTALW

Multiple
alignment
of your
sequences

ETE3	MAFFT	CLUSTALW	PRRN
Help			
General Setting Parameters:			
Output Format: FASTA			
Pairwise Alignment: ☒ FAST/APPROXIMATE ☐ SLOW/ACCURATE			
Enter your sequences (with labels) below (copy & paste): ☐ PROTEIN ☒ DNA			
Support Formats: FASTA (Pearson), NBRF/PIR, EMBL/Swiss Prot, GDE, CLUSTAL, and GCG/MSF			
<pre>>rag1f CAGTACGTTTCTCATTCACAATCATGACCATCAGCATTCCCAACAACAGTGGGTCTGTTAGAATCTTTGAAGAGTCCAAACCAA ACTCTGAACTTTGCTGCAAACCCCTTTGTCTAATGCTTGCTGATGAATCAGACCACGAAACTCTAACAGCCATCTTGAGCCCTT TAATAGCTGAGAGAGAAGCCATGAAATCCAGTGAACATAATGTTGGAAATTGGTGAATCCTCAGAAATTTCAAATTTGTCTTTC GTGGCACAGGGTATGATGAGAAGCTTGTTAGAGAAGTAGAAGGACTGGAAGCCTCAGGCTCTATCTATATTTGTACCCCTCTGTG ATGCTACTCGGCTGGAAGCTTCTCAGAACTTAGTCTTCCATTCTATTACCAGAAGTCATTCTGAGAAGTTACAGCGCTATGAGA CATGGAGAGCCAACCTCACAATGAATCTGCTGACGAGCTTCGTGACAGAGTGAAAGGAGTTTCAGCAAAGCCATTATTGAAA CCCTTCCGTCCATTGATGCCTTACACTGTGACATC >rag1r_c TGTGAGAAAGCATGGCAGTGGTCCACAGTTCCTGAGAAAGCAGTACGTTTCTCATTCACAATCATGACCATCAGCATTCCCAAC AACAGTGGGTCTGTTAGAATCTTTGAAGAGTCCAAACCAAACTCTGAACTTTGCTGCAAACCCCTTTGTCTAATGCTTGCTGAT GAATCAGACCACGAAACTCTAACAGCCATCTTGAGCCCTTTAATAGCTGAGAGAGAAGCCATGAAATCCAGTGAACATAATGTTG GAAATTTGGTGAATCCTCAGAAATTTCAAATTTGTCTTTCGTGGCACAGGGTATGATGAGAAGCTTGTTAGAGAAGTAGAAGGA CTGGAAGCCTCAGGCTCTATCTATATTTGTACCCCTCTGTGATGCTACTCGGCTGGAAGCTTCTCAGAACTTAGTCTTCCATTCT ATTACCAGAAGTCATTCTGAGAAGTTACAGCGCTATGAGACATGGAGAGCCAACCTCACAATGAATCTGCTGACGAGCTTCGT GACAGAGTGAAAGGAGTT]</pre>			
Or give the file name containing your query			
Choose file No file chosen			
Execute Multiple Alignment Reset			

Tool output

Download

Multiple
alignment
of your
sequences

```
>rag1f
```

```
-----CAGTACGTTTCTCATTACACA  
ATCATGACCATCAGCATTCCCAACAACAGTGGGTCTGTTAGAATCTTTGAAGAGTCCAAA  
CCAAACTCTGAACTTTGCTGCAAACCCCTTTGTCTAATGCTTGCTGATGAATCAGACCAC  
GAAACTCTAACAGCCATCTTGAGCCCTTTAATAGCTGAGAGAGAAGCCATGAAATCCAGT  
GAACTAATGTTGGAAATTGGTGGAAATCCTCAGAAATTTCAAATTTGTCTTTTCGTGGCACA  
GGGTATGATGAGAAGCTTGTTAGAGAAGTAGAAGGACTGGAAGCCTCAGGCTCTATCTAT  
ATTTGTACCCTCTGTGATGCTACTCGGCTGGAAGCTTCTCAGAACTTAGTCTTCCATTCT  
ATTACCAGAAGTCATTCTGAGAACTTACAGCGCTATGAGACATGGAGAGCCAACCCCTCAC  
AATGAATCTGCTGACGAGCTTCGTGACAGAGTGAAAGGAGTTTCAGCAAAGCCATTTATT  
GAAACCCCTCCGTCCATTGATGCCTTACACTGTGACATC
```

```
>rag1r_c
```

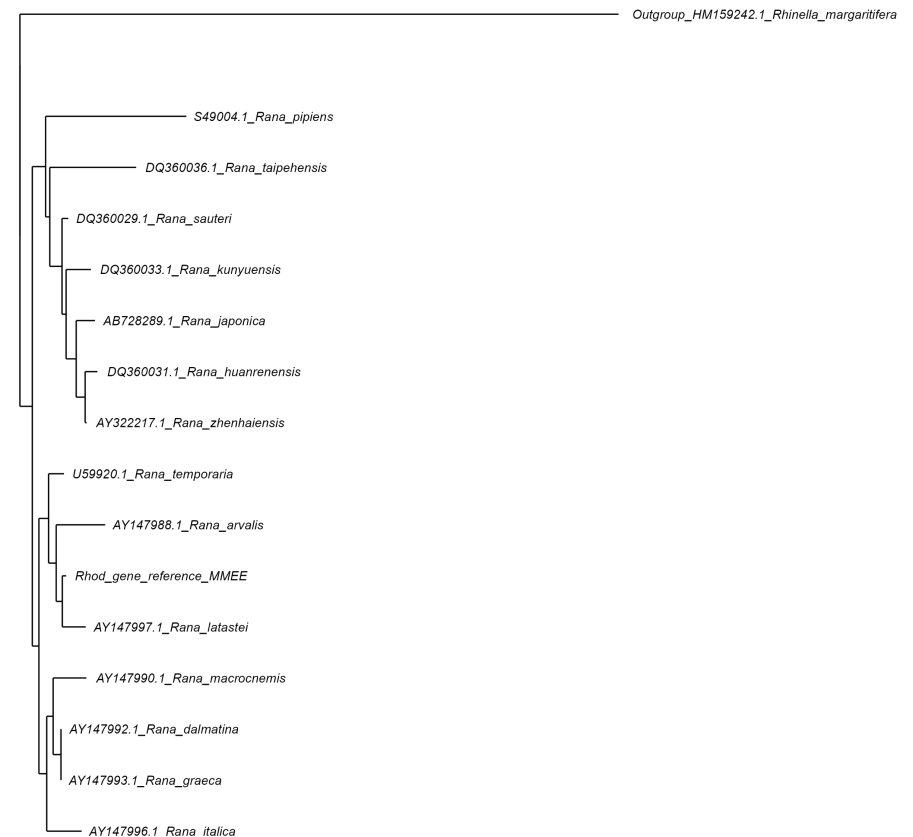
```
TGTGAGAAGCATGGCAGTGGTCCACCAGTTCCTGAGAAAGCAGTACGTTTCTCATTACACA  
ATCATGACCATCAGCATTCCCAACAACAGTGGGTCTGTTAGAATCTTTGAAGAGTCCAAA  
CCAAACTCTGAACTTTGCTGCAAACCCCTTTGTCTAATGCTTGCTGATGAATCAGACCAC  
GAAACTCTAACAGCCATCTTGAGCCCTTTAATAGCTGAGAGAGAAGCCATGAAATCCAGT  
GAACTAATGTTGGAAATTGGTGGAAATCCTCAGAAATTTCAAATTTGTCTTTTCGTGGCACA  
GGGTATGATGAGAAGCTTGTTAGAGAAGTAGAAGGACTGGAAGCCTCAGGCTCTATCTAT  
ATTTGTACCCTCTGTGATGCTACTCGGCTGGAAGCTTCTCAGAACTTAGTCTTCCATTCT  
ATTACCAGAAGTCATTCTGAGAACTTACAGCGCTATGAGACATGGAGAGCCAACCCCTCAC  
AATGAATCTGCTGACGAGCTTCGTGACAGAGTGAAAGGAGTT-----  
-----
```

Multiple alignment of species sequences, including an outgroup

```
1 >Rag1_gene_reference_MMEE
2 GAGAAGCATGGCAGTGGTCCACCAGTTCCTGAGAAAGCAGTACGTTTCTCATTACAATCATGACCATCAGCATTCCCAA
3 >Outgroup_KF666178.1_Rhinella_margaritifera
4 TGAGAAGCATGGCTGTGGACCTGCAGTACCAGAGAAGGCTGTCGGCTTCTCCTTCACCATCATGACCATAAGTGTACCA
5 ACAGCAAAATGGATCTGTAGAATTTTGGAGAGGCTAAACCAATTCTGAACTTGTCTGCAAACCGGTGTGTTAATG
6 CTTGCTGATGAATCAGACCACGAACTCTGACTGCCATCCTGGGTCCTCTGATAGCCGAAAGAGAATCCATGAAATGCAG
7 TGATCTCCTCTTGAAATTTGGAGGGATACGTAGAAGCTTTAAGTTTATCTTCGGGGTACAGGTTATGATGAGAAGTTGG
8 TACGTGAAGTTGAAGGACTTGAAGCCTCAGGTTCTGTCTATATTTGTACTCTTTGTGATGCCACCCGTCTAGAAGCTTCT
9 CAGAACTTGTTTTCCATTCCATTACCAGAAGCCACAGTGAGAACTTGCAACGCTATGAAACATGGAGAGCTAATCCCTA
10 CCATGAATCTGTTGACGACCTTCGTGACAGAGTTAAGGGAGTTTCTGCAAAGCCATTTATTGAAACCTCCCATCCATTG
11 ATGCTTTGCAC
12
13 >KX269563.1_Rana_dalmatina
14 GAGAAGCATGGCAGTGGTCCACCAGTTCCTGAGAAAGCAGTACGTTTCTCATTACAATCATGACCATCAGCATTCCCAA
15 CAACAGTGGGTCTGTTAGAATCTTTGAAGAGTCCAACCAAACCTCTGAACCTTGTCTGCAAACCCCTTTGTCTAATGCTTG
16 CTGATGAATCAGACCACGAACTCTAACAGCCATCTTGAGCCCTTAATAGCTGAGAGAGAAGCCATGAAATCCAGTGAA
17 CTAATGTTGGAAATTTGGTGAATCCTCAGAAATTTCAAATTTGTCTTTCTGCGGCACAGGGTATGATGAGAAGCTTGTTAG
18 AGAAGTAGAAGGACTGGAAGCCTCAGGCTCTATCTATATTTGTACCCTCTGTGATGCTACTCGGCTGGAAGCTTCTCAGA
19 ACTTAGTCTTCCATTCTATTACCAGAAGTCATTCTGAGAACTTACAGCGCTATGAGACATGGAGAGCCAACCTCACAA
20 GAATCTGCTGACGAGCTTCGTGACAGAGTGAAAGGAGTTTCAGCAAAGCCATTTATTGAAACCTTCCGTCCATTGATGC
21 CTTACATTGTGA
22 >KX269559.1_Rana_macrocnemis
23 GAGAAGCATGGCAGTGGTCCACCAGTTCCTGAGAAAGCAGTACGTTTCTCATTACAATCATGACCATCAGCATTCCCAA
24 CAACAGTGGGTCTGTTAGAATCTTTGAAGAGTCCAACCAAACCTCTGAACCTTGTCTGCAAACCCCTTTGTCTAATGCTTG
25 CTGATGAATCAGACCACGAACTCTAACAGCCATCTTGAGCCCTTAATAGCTGAGAGAGAAGCCATGAAATCCAGTGAA
26 CTAATGTTGGAAATTTGGTGAATCCTCAGAAATTTCAAATTTGTCTTTCTGCGGCACAGGGTATGATGAGAAGCTTGTTAG
27 AGAAGTAGAAGGACTGGAAGCYTCAGGCTCTATCTATATTTGTACCCTCTGTGATGCTACTCGGCTGGAAGCTTCTCAGA
28 ACTTAGTCTTCCATTCTATTACCAGAAGTCATTCTGAGAACTTACAGCGCTATGAGACATGGAGAGCCAACCTCACAA
29 GAATCTGCTGACGAGCTTCGTGACAGAGTGAAAGGAGTTTCAGCAAAGCCATTTATTGAAACCTTCCGTCCATTGATGC
30 CTTACATTGTGA
31 >KX269561.1_Rana_temporaria
32 GAGAAGCATGGCAGTGGTCCACCAGTTCCTGAGAAAGCAGTACGTTTCTCATTACAATCATGACCATCAGCATTCCCAA
33 CAACAGTGGGTCTGTTAGAATCTTTGAAGAGTCCAACCAAACCTCTGAACCTTGTCTGCAAACCYCTTTGTCTAATGCTTG
34 CTGATGAATCAGACCACGAACTCTAACAGCCATCTTGAGCCCTTAATAGCTGAGAGAGAAGCCATGAAATCCAGTGAA
35 CTAATGTTGGAAATTTGGTGAATCCTCAGAAATTTCAAATTTGTCTTTCTGCGGCACAGGGTATGATGAGAAGCTTGTTAG
36 AGAAGTAGAAGGACTGGAAGCCTCAGGCTCTATCTATATTTGTACCCTCTGTGATGCTACTCGGCTGGAAGCTTCTCAGA
37 ACTTAGTCTTCCATTCTATTACCAGAAGTCATTCTGAGAACTTACAGCGCTATGAGACATGGAGAGCCAACCTCACAA
38 GAATCTGCTGACGAGCTTCGTGACAGAGTGAAAGGAGTTTCAGCAAAGCCATTTATTGAAACCTTCCRTCCATTGATGC
39 CTTACATTGTGA
```

Phylogenetic analyses of two markers

- Multiple alignment of your sequences with the **Reference sequences** provided
- Using R, compute the distance between these sequences and draw a **Neighbor-Joining** tree





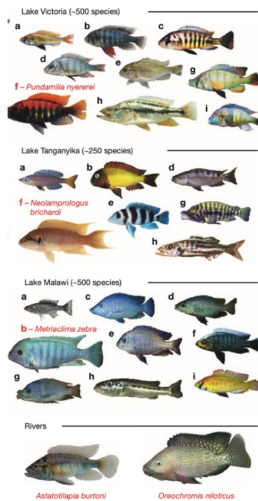
Root the tree: we will also need one or more outgroups, which we know are more distantly related to the other frogs in our tree.

Rhinella margaritifera

Why do we need to include an outgroup?

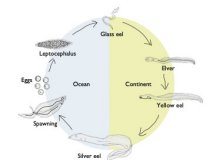
Using molecular approaches to understand the drivers of population divergence and speciation.

Cichlids



Brawand et al., 2014

European eel



Hylid frogs



Simple phylogeny in R

Using R and the packages and tools specified above, load the aligned sequences, compute the distance between these sequences and draw a Neighbour-Joining tree.

You can use *Hyla japonica* as an outgroup to your tree.

Q. What can you say about your sample? Does your sample cluster with other species?

Q. What does this result mean regarding your sample?

Q. What complementary analysis can you think of to better characterize your sample?

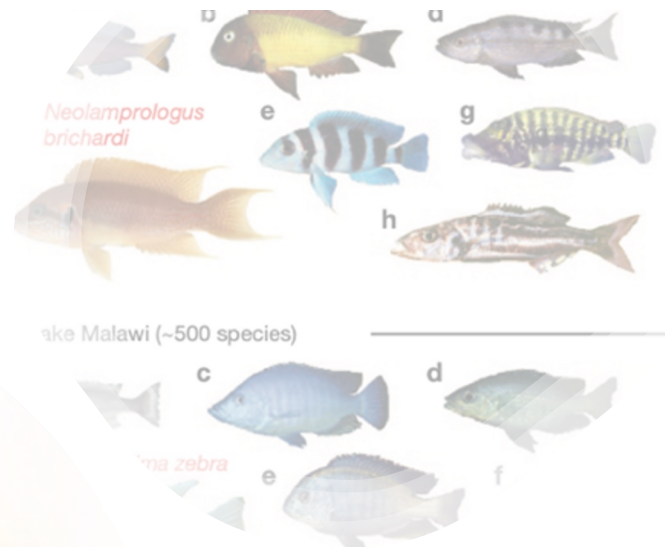
Prof. Ian R. Sanders, Dr. Luca Fumagalli, Prof. Nicolas Salamin

Teaching assistants: Dr. Angélica Pulido, Dr. Giulia Maiello, Marion Nyamari, Hugo Corval, Emelyne Gaudichau, Diego Clavijo.

Q. 1. What can you say about your sample? Does your sample cluster with other species?

Q. 2. What does this result mean regarding your sample?

Q. 3. What complementary analysis can you think of to better characterize you sample?



Project 2

Cryptic speciation in tree frogs

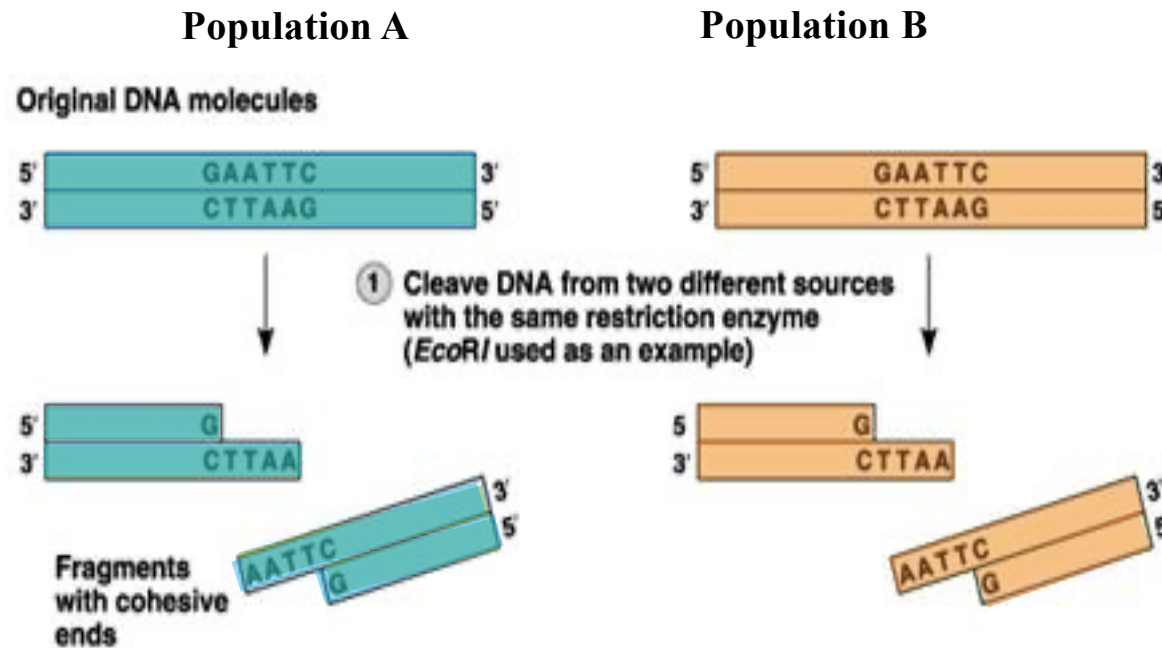
Allopatric divergence – are they different species?

RAD sequencing

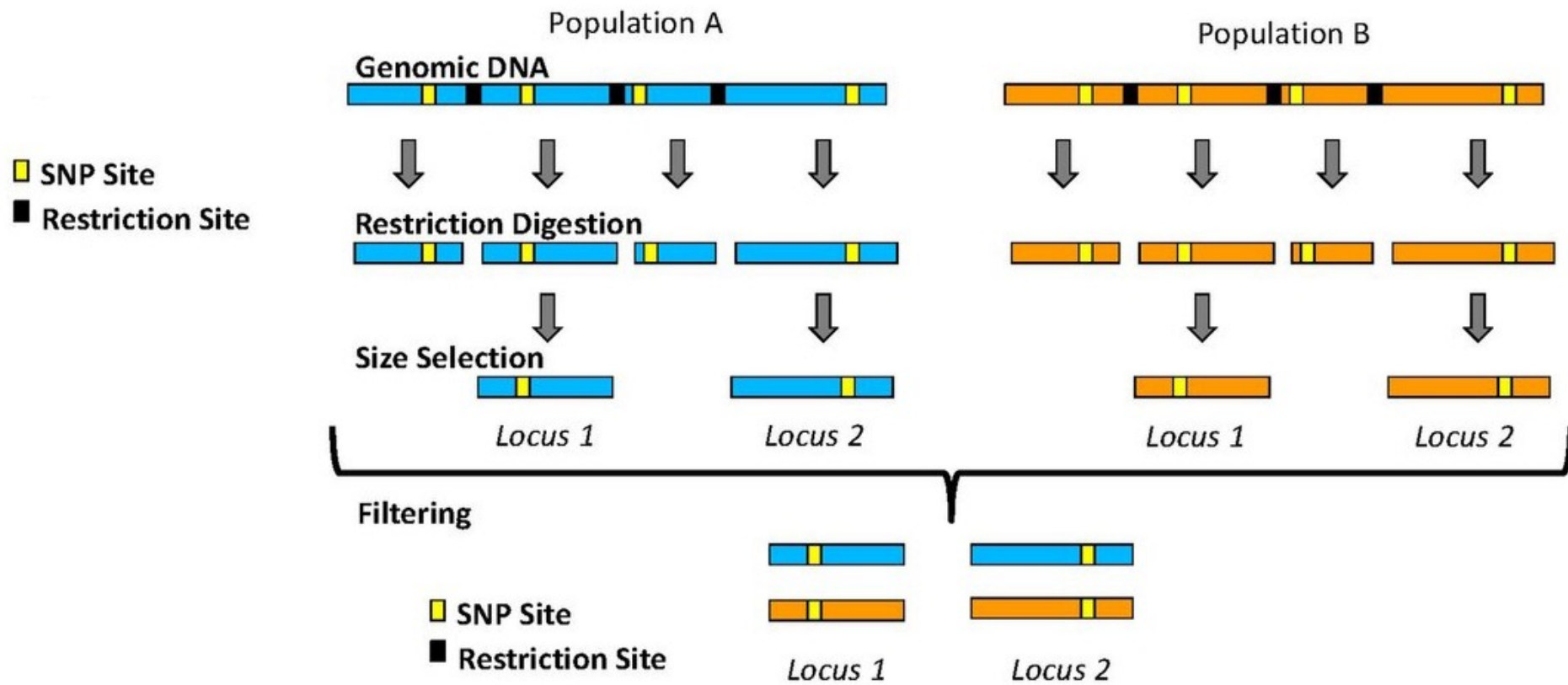
RAD-seq:

Restriction-site Associated DNA (RAD) sequencing

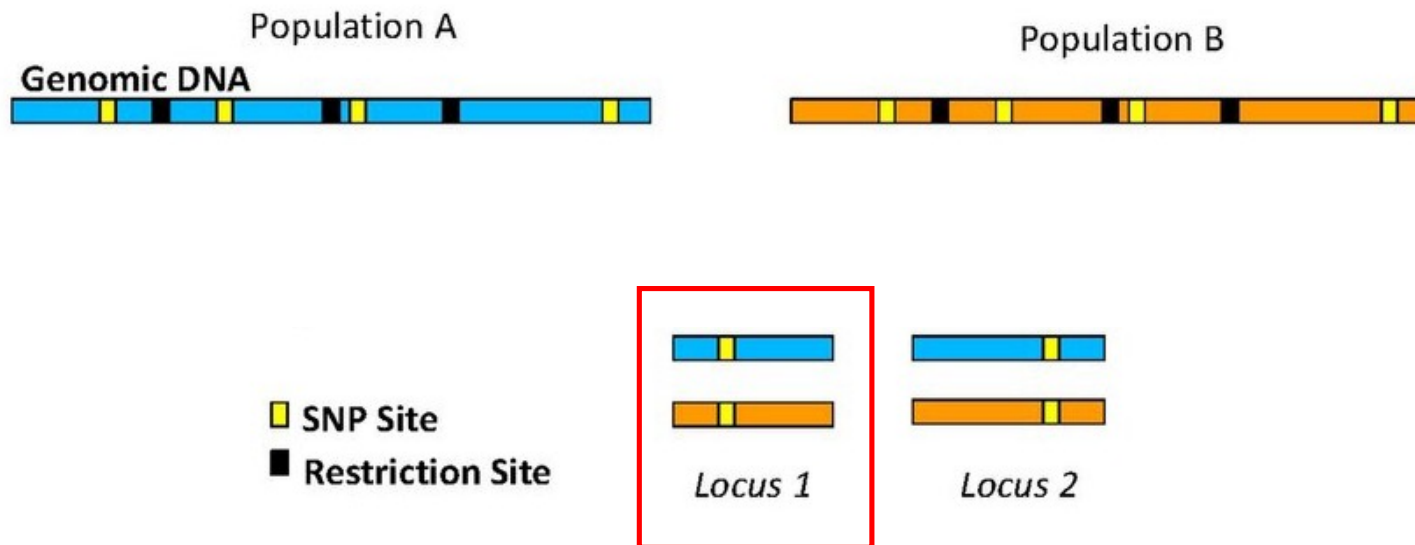
Reduced representation sequencing method



RAD-seq



SNP: Single Nucleotide Polymorphism



SNP: Single Nucleotide Polymorphism

■ SNP Site
■ Restriction Site

Population A



Population B



Locus 1

Individual 1

Maternal ...CGATATTCC**T**ATCGAATGTC...

Paternal ...CGATATTCC**T**ATCGAATGTC...

Individual 2

Maternal ...CGATATTCC**T**ATCGAATGTC...

Paternal ...CGATATTCC**T**ATCGAATGTC...

Individual 3

Maternal ...CGATATTCC**A**ATCGAATGTC...

Paternal ...CGATATTCC**A**ATCGAATGTC...

Individual 4

Maternal ...CGATATTCC**A**ATCGAATGTC...

Paternal ...CGATATTCC**A**ATCGAATGTC...

SNP: Single Nucleotide Polymorphism

Most common type of **genetic variation** and can originate by **mutations**.
They can accumulate over time and spread across individuals in a population.

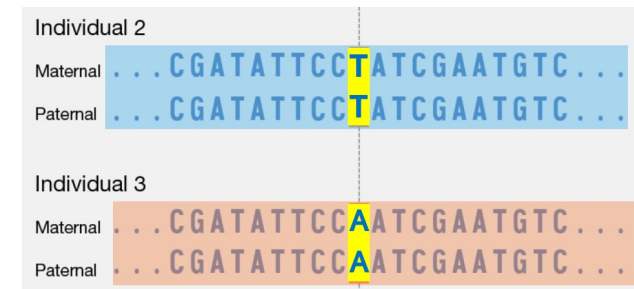


By comparing SNP patterns across different species or populations, we can:

- Build phylogenetic trees between populations and species
- Understanding Population Evolution: e.g hybridization, bottle necks, migration patterns
- Identifying Selection Pressure: Type of selection, Adaptive traits (SNPs associated with traits)

The Variant Call Format (VCF)

tab-delimited text file that store genetic variations, such as **single nucleotide polymorphisms (SNPs)**



VCF

```
##fileformat=VCFv4.2
##contig=<ID=2,length=51304566>
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
```

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2	SAMPLE3	SAMPLE4
2	81170	.	C	T	.	.	AC=9;AN=7424	GT:DP:GQ	0/0:4:12	0/0:3:9	0/1:1:3	0/1:9:24
2	81171	.	G	A	.	.	AC=6;AN=7446	GT:DP:GQ	0/1:4:12	0/0:3:9	0/0:1:3	0/0:9:24
2	81182	.	A	G	.	.	AC=5;AN=7506	GT:DP:GQ	0/0:5:15	0/0:4:12	0/0:5:15	0/0:9:24
2	81204	.	T	G	.	.	AC=2;AN=7542	GT:DP:GQ	1/0:5:15	0/0:9:27	0/0:10:30	0/0:15:39

Why RAD-seq?

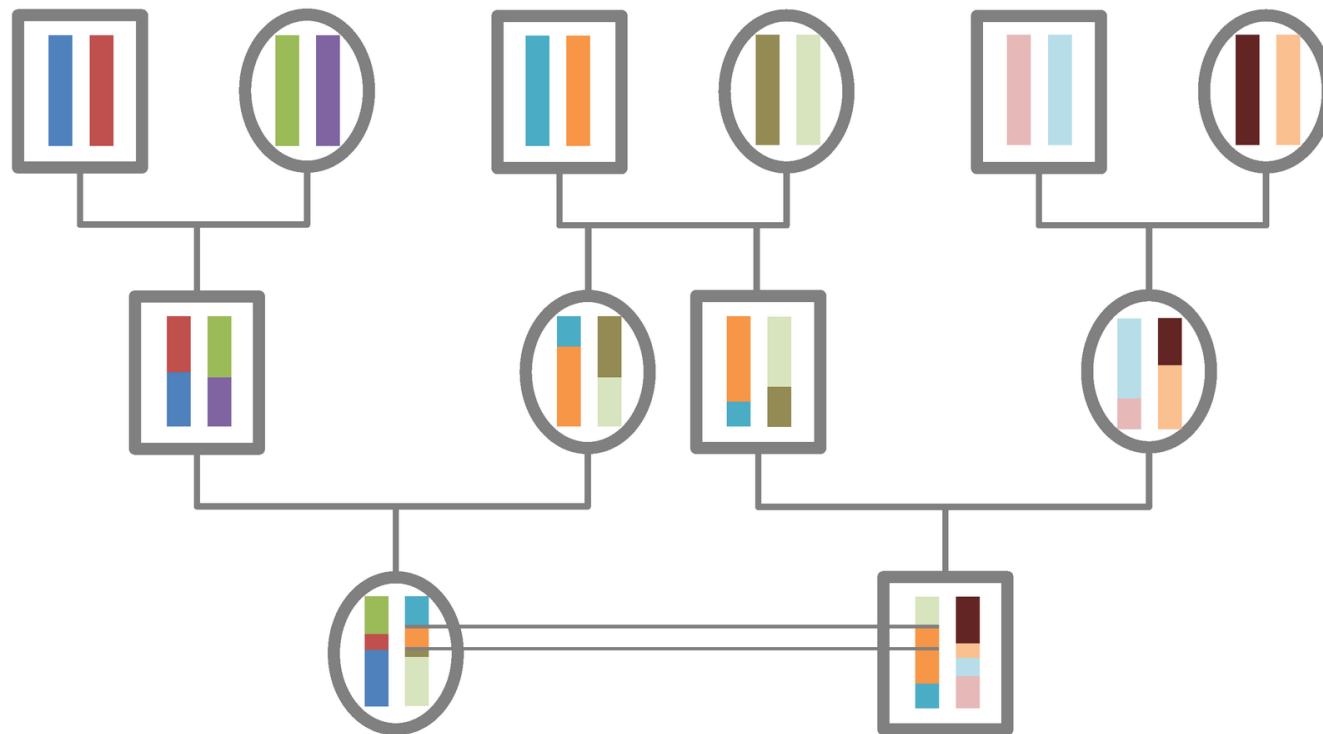
... Whole-genome data

... Recombination

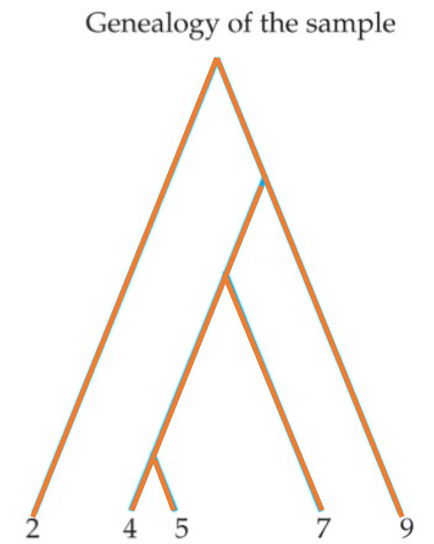
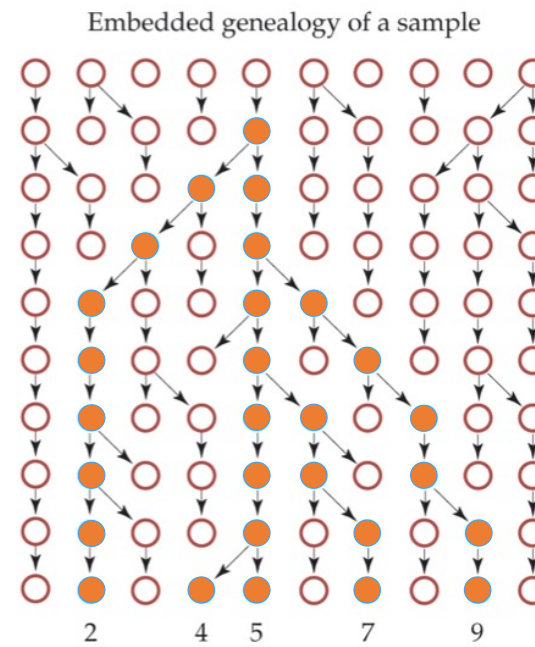
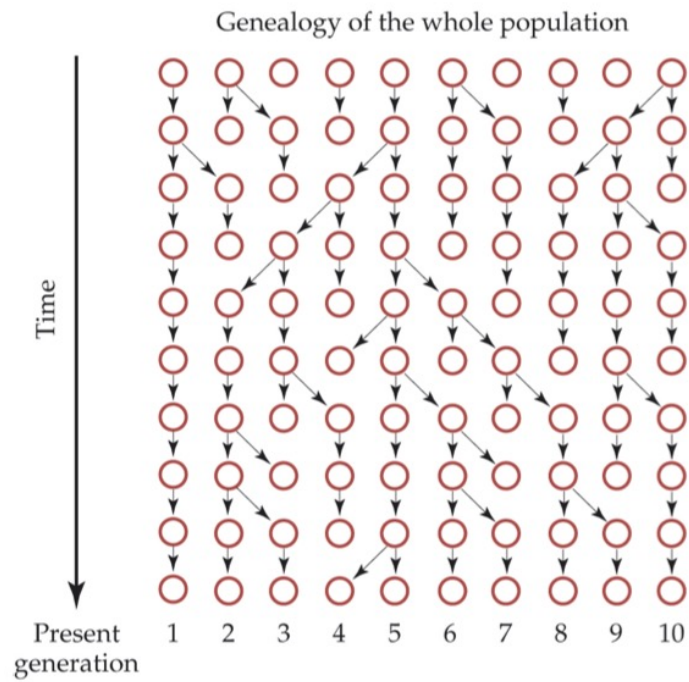


Why RAD-seq?

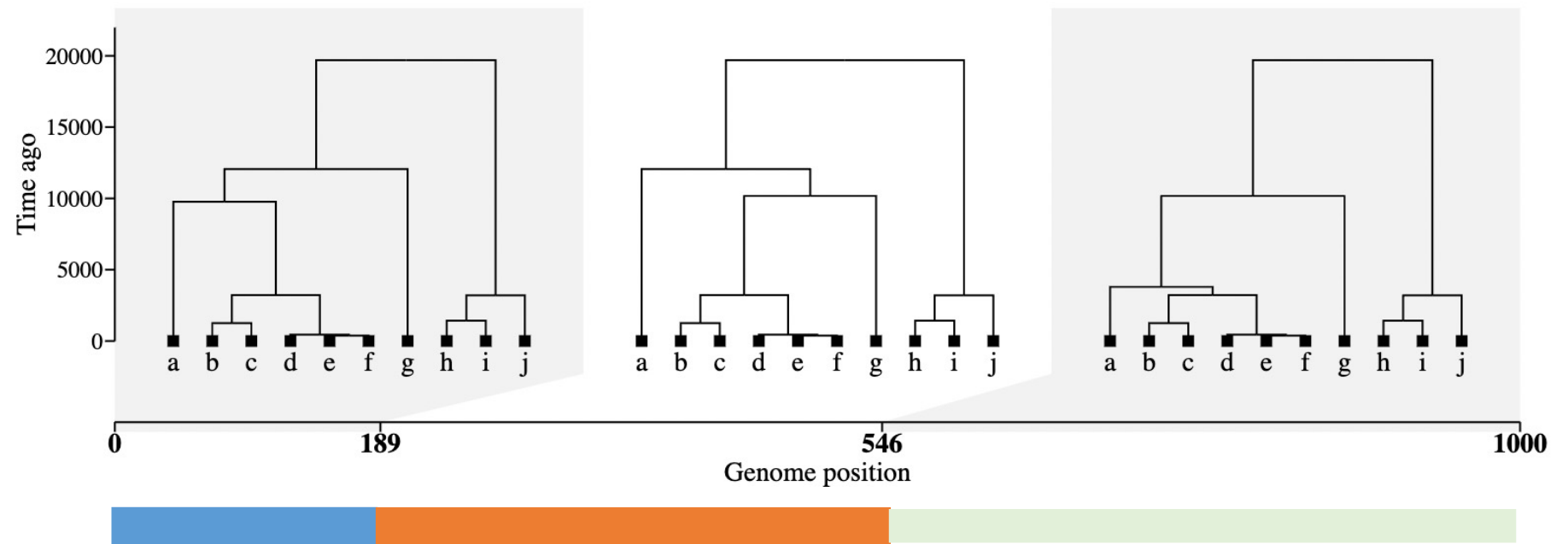
Recombination

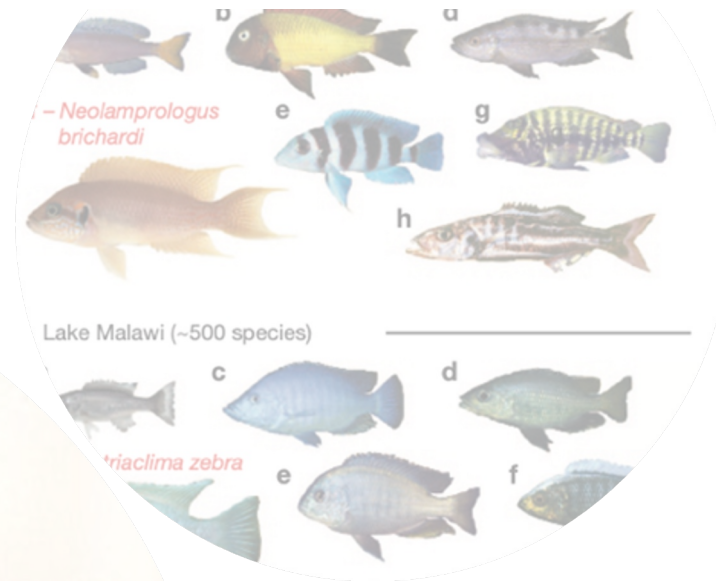
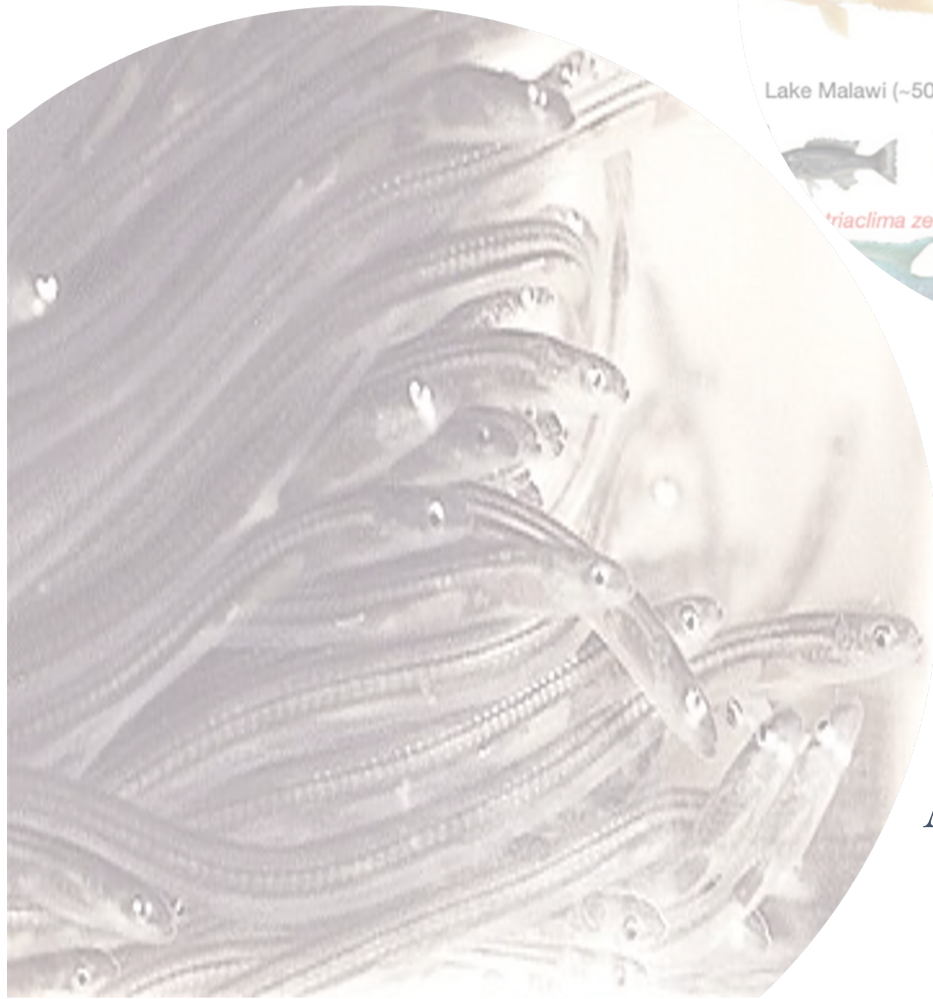


The Species Coalescent



The Species Coalescent across the genome



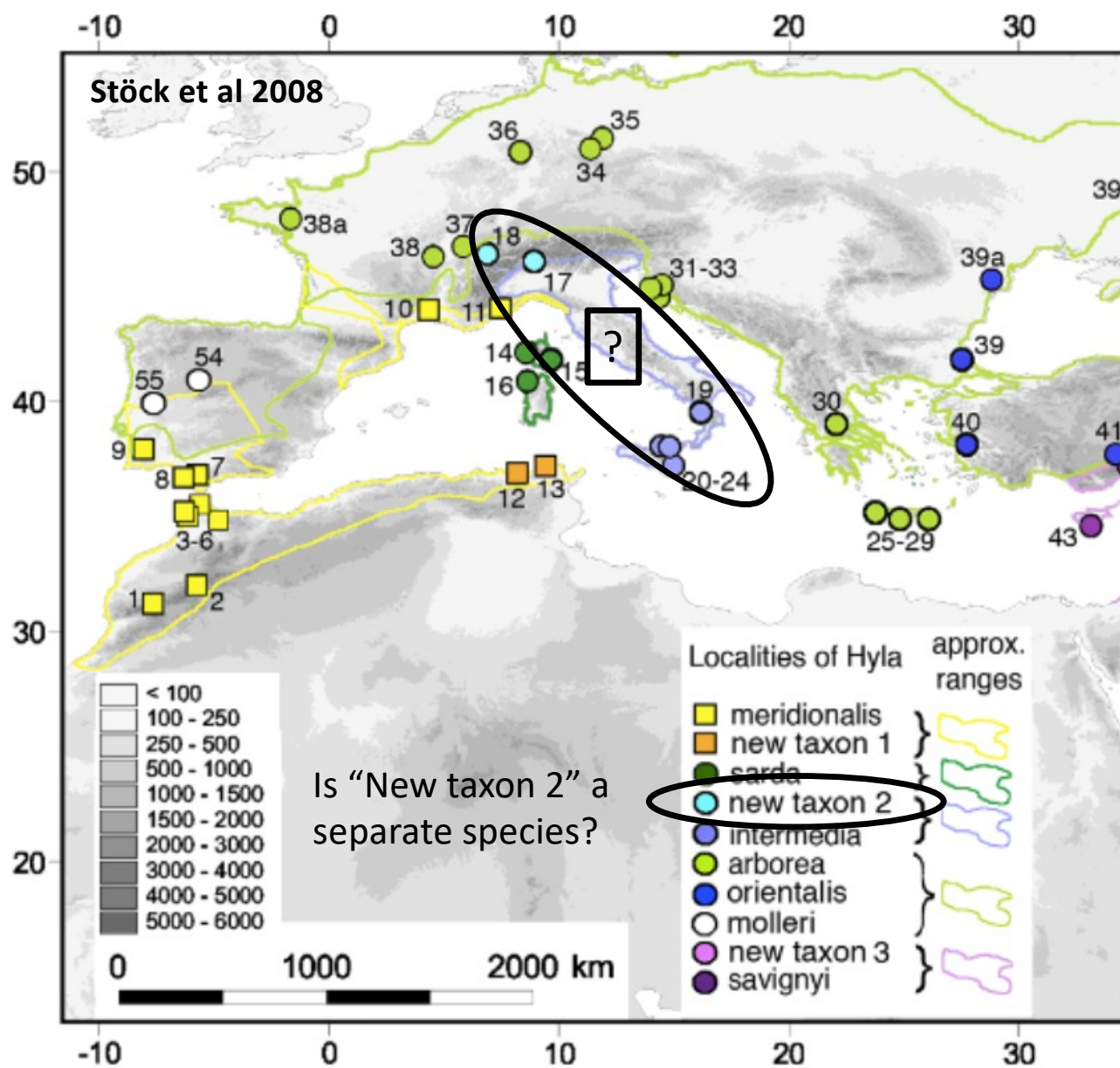


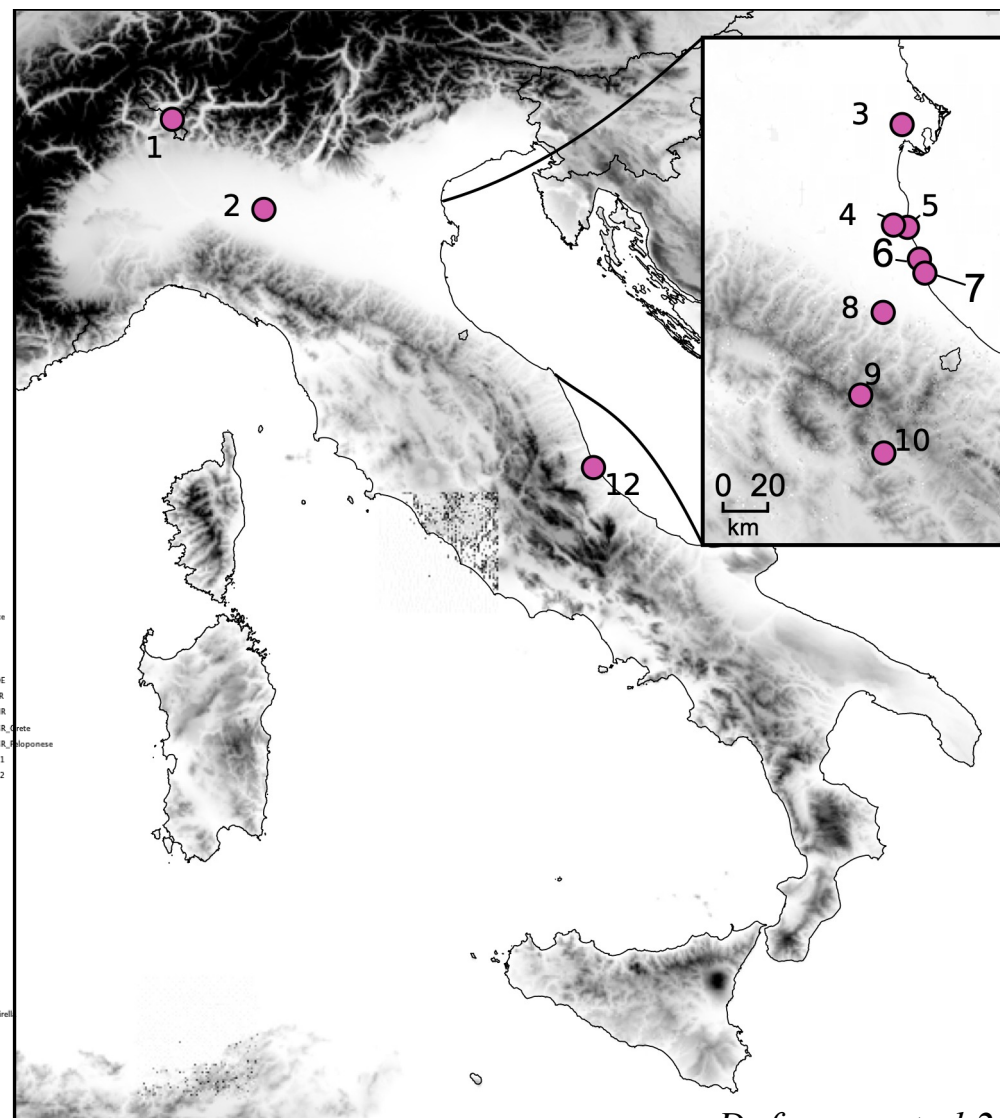
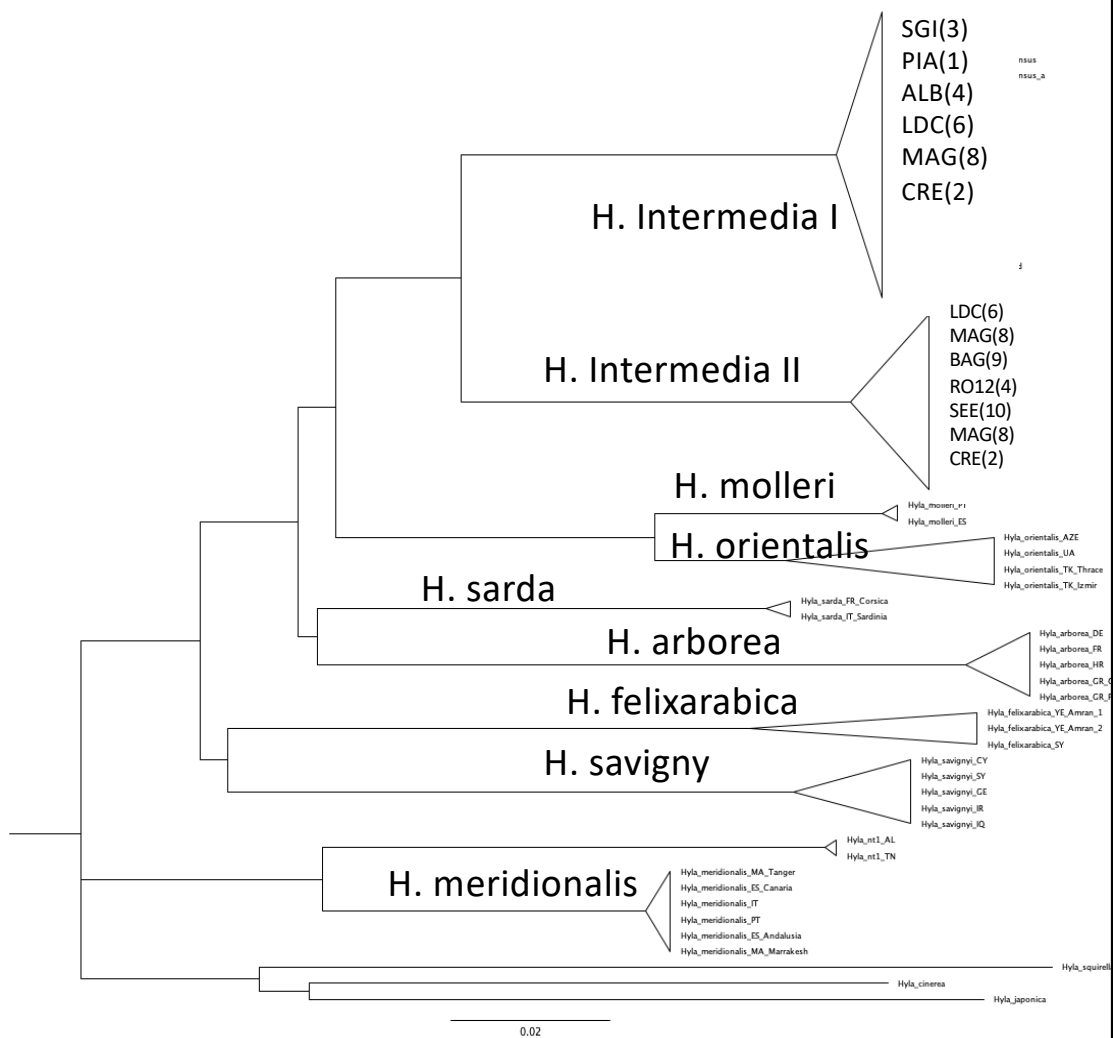
Project 2

Cryptic speciation in tree frogs

Allopatric divergence – are they different species?

RAD sequencing





Dufresnes et al 2018

FSTAT

read.fstat.data {hierfstat}

hyla_genind										
	Pop	100057	100057.1	100057.2	100065	100216	100224	100224.1	100249	100397
1	1	22	22	column 4: numeric with range 10 - 34	33	31	22	11	24	
2	1	22	22	31	22	31	31	42	11	22
3	1	24	22	33	22	33	31	42	11	22
4	1	22	22	33	22	31	11	22	11	22
5	1	44	22	33	22	33	31	4		
6	1	22	22	33	22	33	31	4		
7	1	22	22	33	22	33	33	4		
8	1	22	22	31	22	33	31	4		
9	1	22	22	33	22	33	11	2		
10	1	22	22	11	21	11	33	4		
11	1	22	24	31	22	11	33	4		
12	1	22	22	31	22	11	33	4		
13	1	22	22	31	22	11	33	4		

Individual 2	
Maternal	...CGATATTCC T ATCGAATGTC...
Paternal	...CGATATTCC T ATCGAATGTC...
Individual 3	
Maternal	...CGATATTCC A ATCGAATGTC...
Paternal	...CGATATTCC A ATCGAATGTC...

EnvironmentHistoryConnectionsTutorial

Import Dataset30 MiB

RGlobal Environment

hyla_genind120 obs. of 2431 variables

hyla_genind_ade...Large genind (3.6 MB)

hyla_genpopLarge genpop (1.5 MB)

Hyla_lat_long12 obs. of 3 variables

hyla_summ_hierLarge basic.stats (7 elements, 2.9 MB)

IBDList of 7

pcaSNPLarge indpca (3 elements, 15 MB)

Pop120 obs. of 1 variable

popcodes120 obs. of 2 variables

ppfst.hyla_genind num [1:12, 1:12] NA 0.106 0.128 0.147 0.148 ...

FilesPlotsPackagesHelpViewerPresentation

read.fstat.data

R: Reads data from a FSTAT fileFind in Topic

Q.1. What caused the divergence between these lineages / species?

Q. 2. Are they one species, or two?



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Short Communication

Mitochondrial and nuclear phylogeny of circum-Mediterranean tree frogs from the *Hyla arborea* group

Matthias Stöck^{a,*}, Sylvain Dubey^{a,b}, Cornelya Klütsch^{c,d}, Spartak N. Litvinchuk^e, Ulrich Scheidt^f, Nicolas Perrin^a

 **frontiers**
in Ecology and Evolution

ORIGINAL RESEARCH
published: 02 October 2018
doi: 10.3389/fevo.2018.00144

Genomic Evidence for Cryptic Speciation in Tree Frogs From the Apennine Peninsula, With Description of *Hyla perrini* sp. nov



Christophe Dufresnes^{1,2,3*}, Glib Mazepa^{1,4}, Nicolas Rodrigues¹, Alan Brelford^{1,5}, Spartak N. Litvinchuk⁶, Roberto Sermier¹, Guillaume Lavanchy¹, Caroline Betto-Colliard¹, Olivier Blaser¹, Amaël Borzée⁷, Elisa Cavoto¹, Guillaume Fabre¹, Karim Ghali¹, Christine Gossen¹, Agnes Horn¹, Julien Leuenberger¹, Barret C. Phillips¹, Paul A. Saunders¹, Romain Savary¹, Tiziano Maddalena⁸, Matthias Stöck⁹, Sylvain Dubey^{1,3}, Daniele Canestrelli¹⁰ and Daniel L. Jeffries¹

<https://github.com/Angelica-Pulido/MMEE-2026/>

Angelica-Pulido / MMEE-2026

Q Type to search

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Files

main

Go to file

> 1.Frogs_Sanger

> 2.Frogs_RADseq

> Data

.DS_Store

2.0.README_Frogs_RADseq.md

> 3.Cichlids

> 4.Eels

.DS_Store

MMEE_2026_Manual.pdf

README.md

Schedule.png

MMEE-2026 / 2.Frogs_RADseq / 2.0.README_Frogs_RADseq.md

Angelica-Pulido Update and rename 1.0.README_Frogs_RADseq.md to 2.0.README_Frogs_RADs... b7f58cf · last year Histor

Preview Code Blame 19 lines (15 loc) · 796 Bytes

Raw

+

+

+

+

+

+

Project: 2.Frogs_RADSeq/

The folders and files for this project are as follow:

Data/RADphylo/:

You can use this file to check plot a NJ tree of Hyla individuals. But this is optional, you don't need this file

Data/RADpopgen/:

- hyla_FSTAT.dat
Data for population analyses in fstat format
- populations_numeric_codes.txt
File with a numeric code for each population.
You have to use this file to change the population column in the R object created when you read the hyla_FSTAT

Data/RADstructure/:

- Hyla.struct.geno
data for Structure-like analyses in geno format, ready for sNMF analyses.

```

### Create a map
library(maps)      # Provides functions that let us plot the maps
library(mapdata)   # Contains the hi-resolution points that mark out the countries.

map('worldHires', xlim = c(3,18), ylim = c(35,48))
points(Hyla_lat_long$Long, Hyla_lat_long$Lat,

```

HIERFSTAT:

HIERFSTAT is a package developed for R to study population structure with the help of genetic markers. Wright's F-statistics are commonly used to determine population structure and differentiation. HIERFSTAT allows you to estimate F-statistics and many other basic statistics important for the analyses of population-level genetic data.

Some commands and functions from HIERFSTAT that you might find useful are listed below but see here <https://www.rdocumentation.org/packages/hierfstat> and help pages of each function for more detailed information:

read.fstat.data(): import data formatted as a .dat file (the .dat file is the format used by the software FSTAT). The data will be read in as a dataframe.

basic.stats(): calculate basic statistics such as expected heterozygosity, observed heterozygosity, F_{IS} , etc for each population and each locus.

pairwise.neifst(): to calculate population pairwise F_{st}

indpca(): to carry out a PCA on the centred matrix of individual allele frequencies

Population genetics of *Hylid* frogs in R

With the R tools mentioned above, and others you might find, explore and analyse the frog SNP dataset, produced with the RADseq approach. The data set can be found here:

`./MolGen-2026/2.Frogs_RADseq/Data/RADpopgen/hyla_FSTAT.dat`

We encourage you to explore the data thoroughly:

- Look at within population statistics, like observed and expected heterozygosity, inbreeding (F_{IS}) and expectations of Hardy Weinberg.
- Look at between population statistics - pairwise F_{ST} between populations, patterns of isolation by distance (i.e. a correlation between physical distance and genetic distance between populations (see also Mantel test)).
- Look at population clustering (i.e. are some populations more closely related to each other than to any others?) using a PCA.

Remember to keep the biological question in mind all the way through - are these frogs one species, or two?

Q. What do these calculations tell you about the divergence between populations?

Q. Are there some populations which are more divergent than others?

Q. Do the populations fall into discrete “groups” which might be considered as different species?

Q. What was the most likely evolutionary history of these species? Has the hybrid zone always been there? How did reproductive isolation occur?

Within population statistics:

- observed and expected heterozygosity
- inbreeding (FIS)
- conclude under expectations of Hardy Weinberg

Between population statistics:

- pairwise FST
- isolation by distance (IBD): physical distance and genetic distance
 - Mantel test

Population clustering:

- PCA
- Structure plot

HIERFSTAT:

HIERFSTAT is a package developed for R to analyse population-level genetic data. Wright's F-statistics are commonly used in such analyses. HIERFSTAT allows you to estimate F-statistics from population-level genetic data.

Some commands and functions from HIERFSTAT are listed here <https://www.rdocumentation.org/packages/HIERFSTAT> for detailed information:

`read.fstat.data()`: import data from FSTAT software. The data will be read in a `basic.stats()` object. `basic.stats()`: calculate basic statistics such as heterozygosity, F_{IS} , etc for each population. `pairwise.neifst()`: to calculate pairwise F_{ST} values. `indpca()`: to carry out a PCA on the centred and scaled data.

```
### Create a map
library(maps)           # Provides functions that let us plot the maps
library(mapdata)        # Contains the hi-resolution points that mark out the countries
```

ADEGENET:

ADEGENET and HIERFSTAT are built using the same underlying principles, therefore, a lot of overlap between them. ADEGENET also uses the Genind object to store allele frequencies, F_{ST} etc. However, ADEGENET is more focused on spatial genetics, for example examining Isolation by Distance (IBD).

Some functions that you will find useful are listed here: <http://adegenet.r-forge.r-project.org/>. The PCA and IBD sections which should provide a good starting point.

`read.fstat()`: “Genind” object - a specific type of object, you can access slots via `slot()`. `pop.names()` show you the information in the Populations slot. `genind2genpop()`: transform a genind object into a genpop object. `dist.genpop()`: Create a Euclidean distance matrix. `dist()`: Create a Euclidean distance matrix. `mantel.randtest()`: Perform a mantel test on the distances (such as pairwise F_{ST}) and pairwise geographical distances.

LEA:

LEA is an R package dedicated to landscape genetics analyses of population structure and genomic data for estimating ancestry coefficients from ancestral populations (snmf, pca).

Some commands and functions that are listed here: <https://www.rdocumentation.org/packages/LEA> for information:

`snmf()`: estimates admixture coefficients using different algorithms, and provides STRUCTURE-like plots. `cross.entropy()`: Return the cross-entropy for the populations.

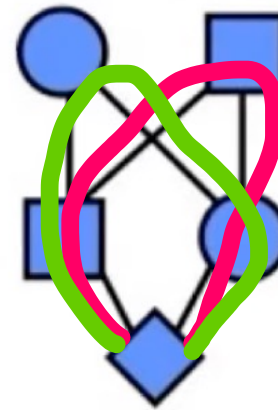
Inbreeding Coefficient (F)

- Measures the **probability** that two alleles at a locus in an individual are identical by descent (IBD).
- IBD: meaning they are copies of the same allele from a common ancestor.
- The level of inbreeding of an offspring is determined by the closeness of the pedigree relationship between the parents

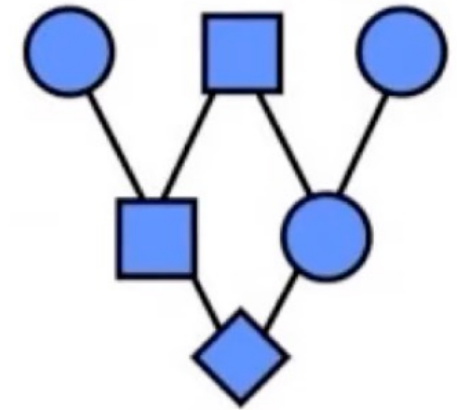
$$F = \sum \left(\frac{1}{2}\right)^N$$

FROH: The genomic inbreeding coefficient calculated by identifying **runs of homozygosity (ROH)**—segments of DNA where both copies of a chromosome are identical by descent from a common ancestor.

Full siblings



Half siblings



$$F = \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3$$

$$\frac{1}{8} + \frac{1}{8} = \frac{1}{4} = 0.25$$

Inbreeding Coefficient within Subpopulations (F_{IS})

Wright's F-statistics and measures the reduction in heterozygosity of an individual relative to its subpopulation due to non-random mating.

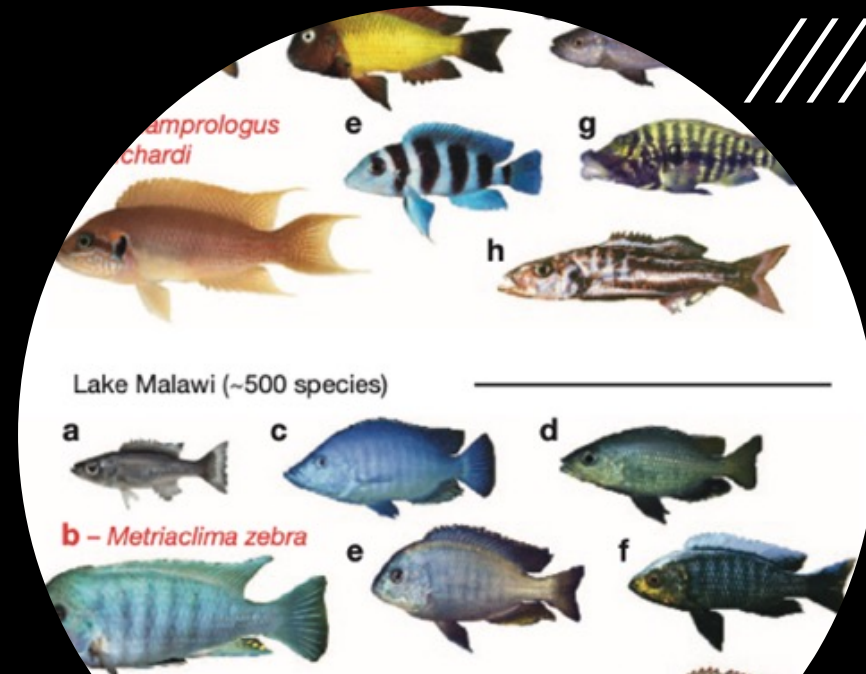
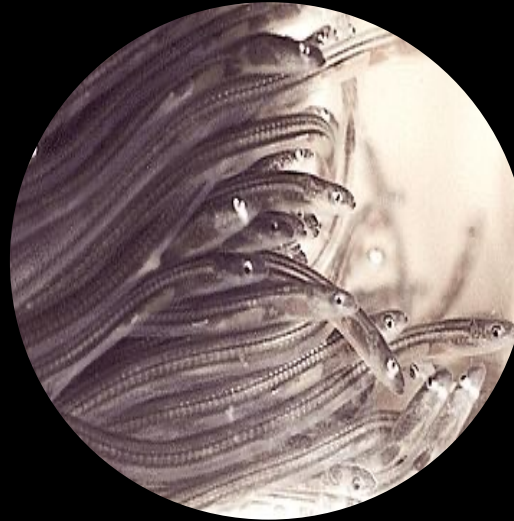
$$F_{IS} = \frac{H_{\text{exp}} - H_{\text{obs}}}{H_{\text{exp}}}$$

$$F_{IS} > 0$$

$$F_{IS} = 0$$

$$F_{IS} < 0$$

Molecular Methods in Ecology and Evolution 2026



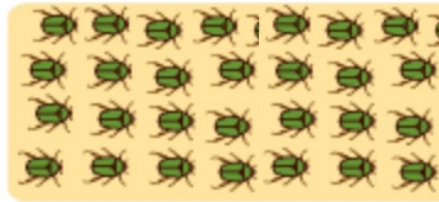


Project 3

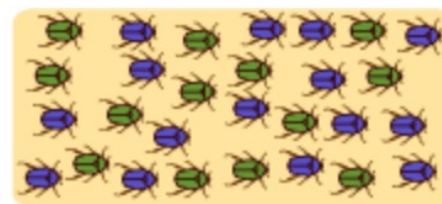
Divergence between Lake Malawi cichlids.

Ecological speciation in sympatry

Initial population



Geographical / Allopatric
divergence



Ecological / sympatric
divergence



The genomic basis of cichlid fish adaptation within the deepwater “twilight zone” of Lake Malawi

Christoph Hahn,^{1,2,3} Martin J Genner,⁴ George F Turner,⁵ and Domino A Joyce¹ 

¹*Evolutionary and Environmental Genomics Group (@EvoHull), School of Environmental Sciences, University of Hull, Hull HU5 7RX, United Kingdom*

²*Institute of Zoology, University of Graz, A-8010 Graz, Austria*

³*E-mail: christoph.hahn@uni-graz.at*

⁴*School of Biological Sciences, University of Bristol, Bristol Life Sciences Building, 24 Tyndall Avenue, Bristol BS8 1TQ, United Kingdom*

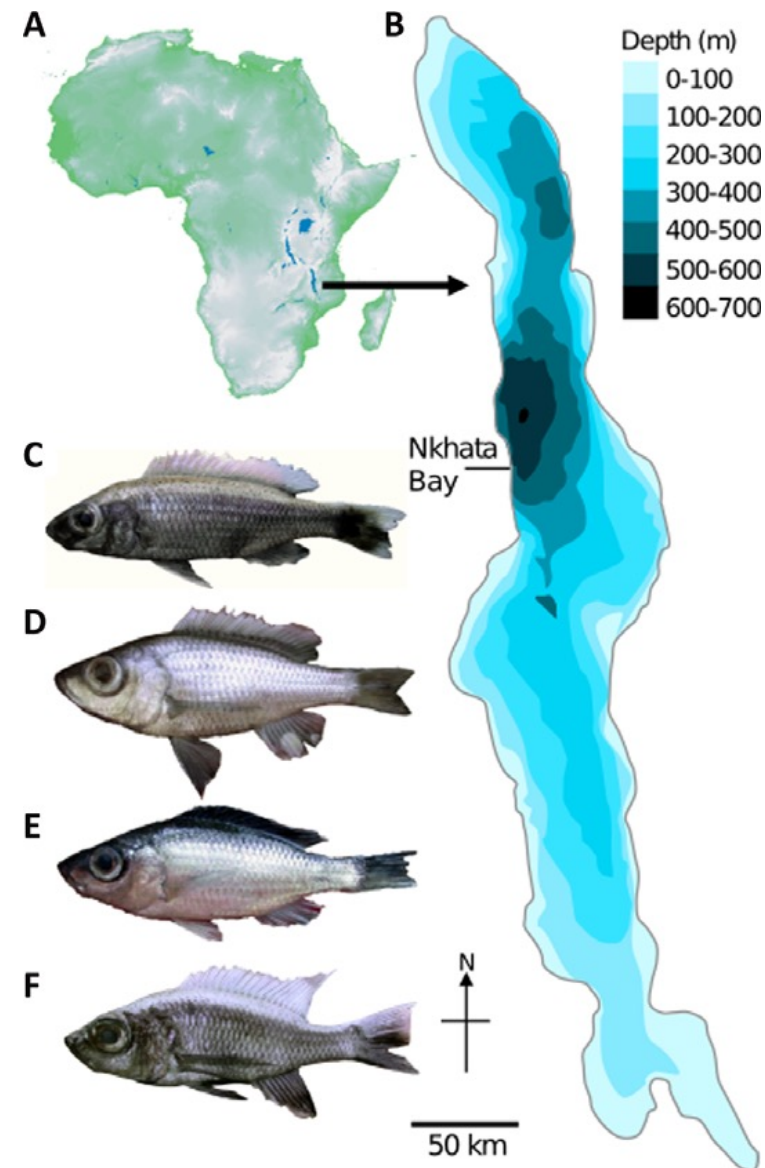
⁵*School of Biological Sciences, Bangor University, Bangor, Gwynedd LL57 2UW, Wales, United Kingdom*

Cichlids sympatric speciation:

Maps of Africa (topographic) and Lake Malawi indicating the sampling location Nkhata Bay.

Diplotaxodon spp

- (C) *D. “limnothrissa black pelvic”*;
- (D) *D. “macrops offshore”*;
- (E) *D. “macrops black dorsal”*;
- (F) *D. “macrops ngulube.”*



Cichlids sympatric speciation:

Deep water environment characteristics:

- High hydrostatic pressure
- Low light
- Low dissolved oxygen

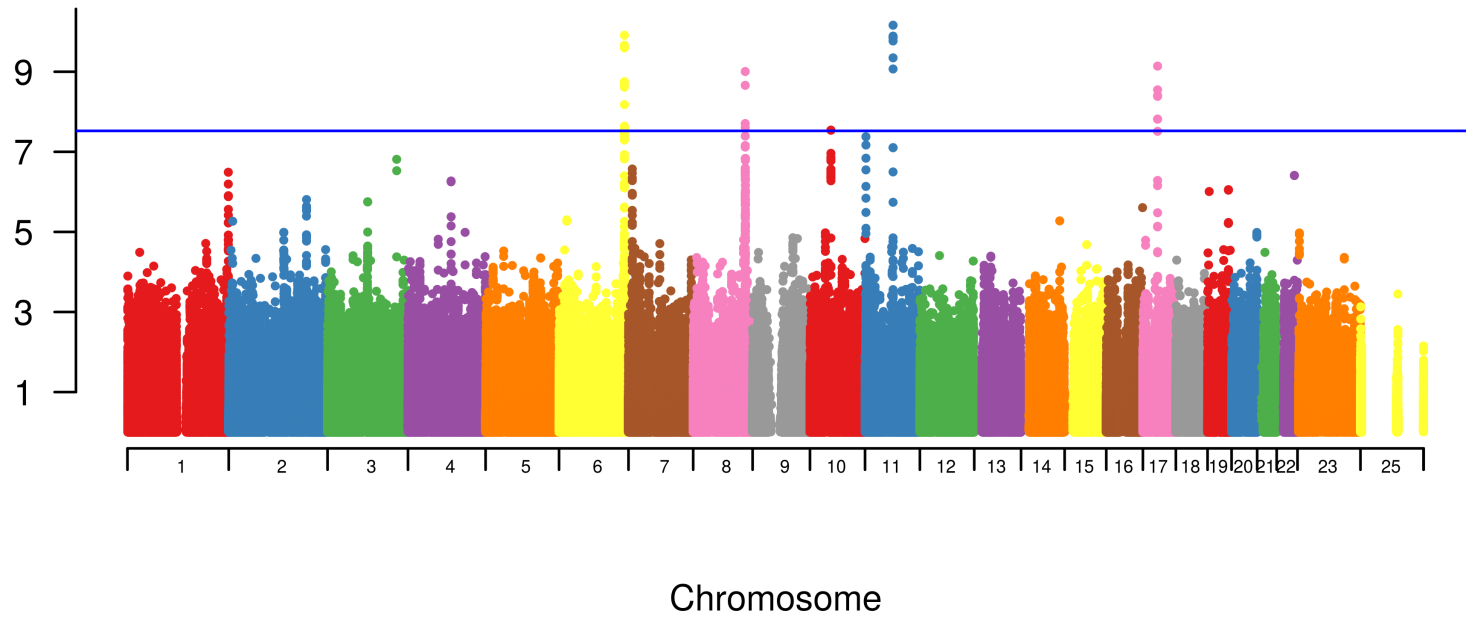
Q.1. What can be driving this sympatric divergence?

Q. 2. Are there some populations which are more diverged than others?
If so, why?

Q.3. Can you identify candidate loci under selection?

Plot genomic data:

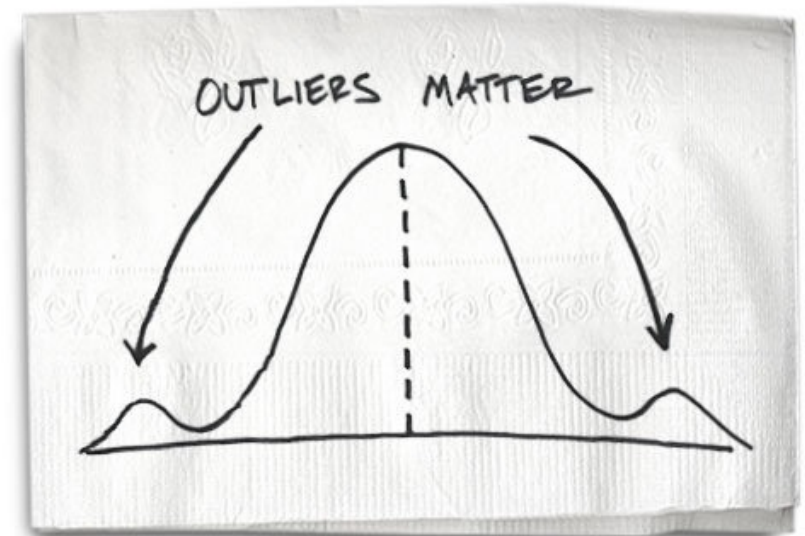
Manhattan plot



Outlier detection:

Causes of outliers:

- Data entry and measurements errors
- Experimental
- Data processing errors
- Natural (not an error!!!! novelties in data)



<https://github.com/Angelica-Pulido/MMEE-2026/>

Angelica-Pulido / MMEE-2026

Q Type to search

+ 🔍

<> Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Files

main

Go to file

> 1.Frogs_Sanger

> 2.Frogs_RADseq

> 3.Cichlids

> Data

3.0.README_Cichlids.md

> 4.Eels

.DS_Store

MMEE_2026_Manual.pdf

README.md

Schedule.png

MMEE-2026 / 3.Cichlids / 3.0.README_Cichlids.md

Angelica-Pulido Update 3.0.README_Cichlids.md 9f27ec1 · 11 months ago

Preview Code Blame 24 lines (15 loc) · 733 Bytes

Raw

Project: 3.Cichlids/

The files and folders for this project are as follow:

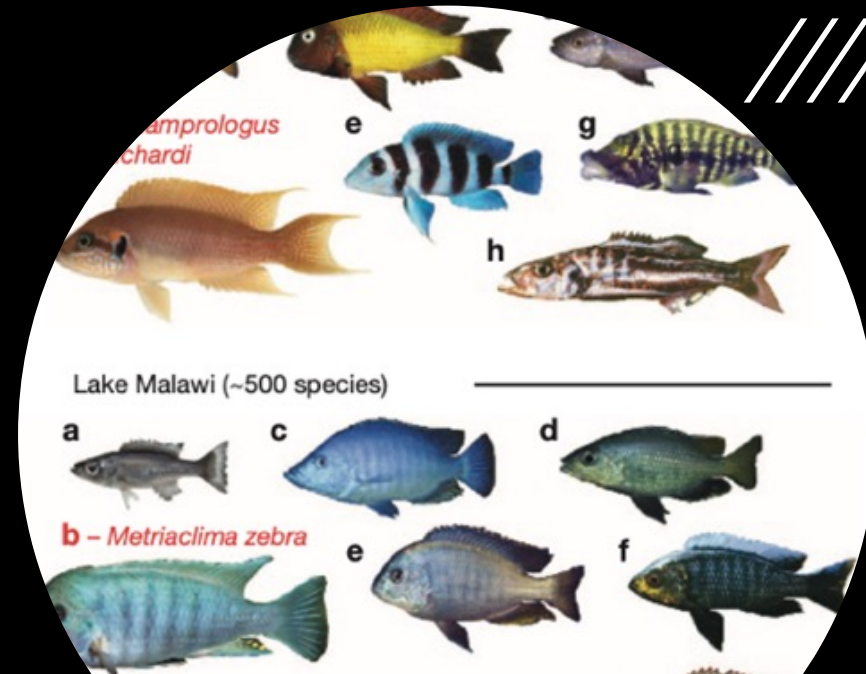
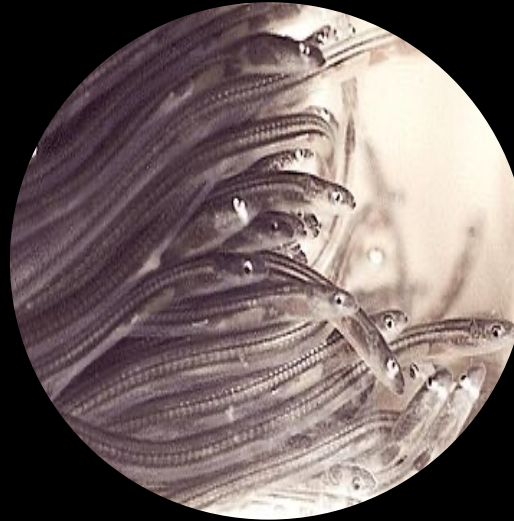
Data/:

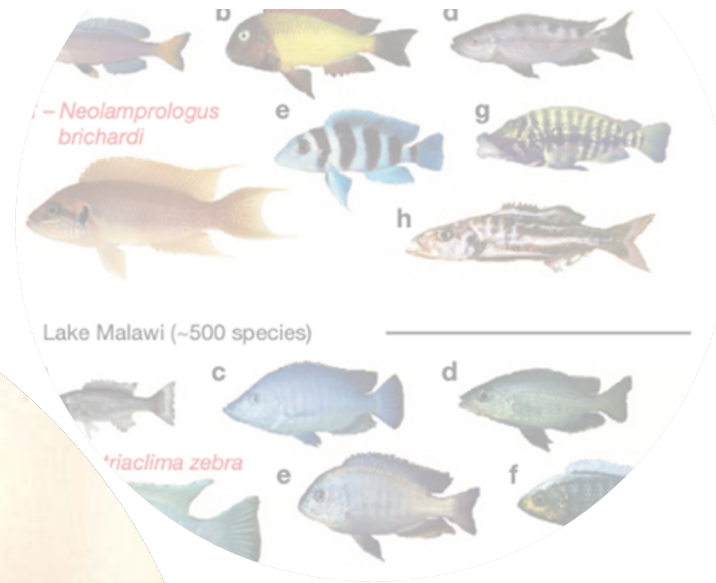
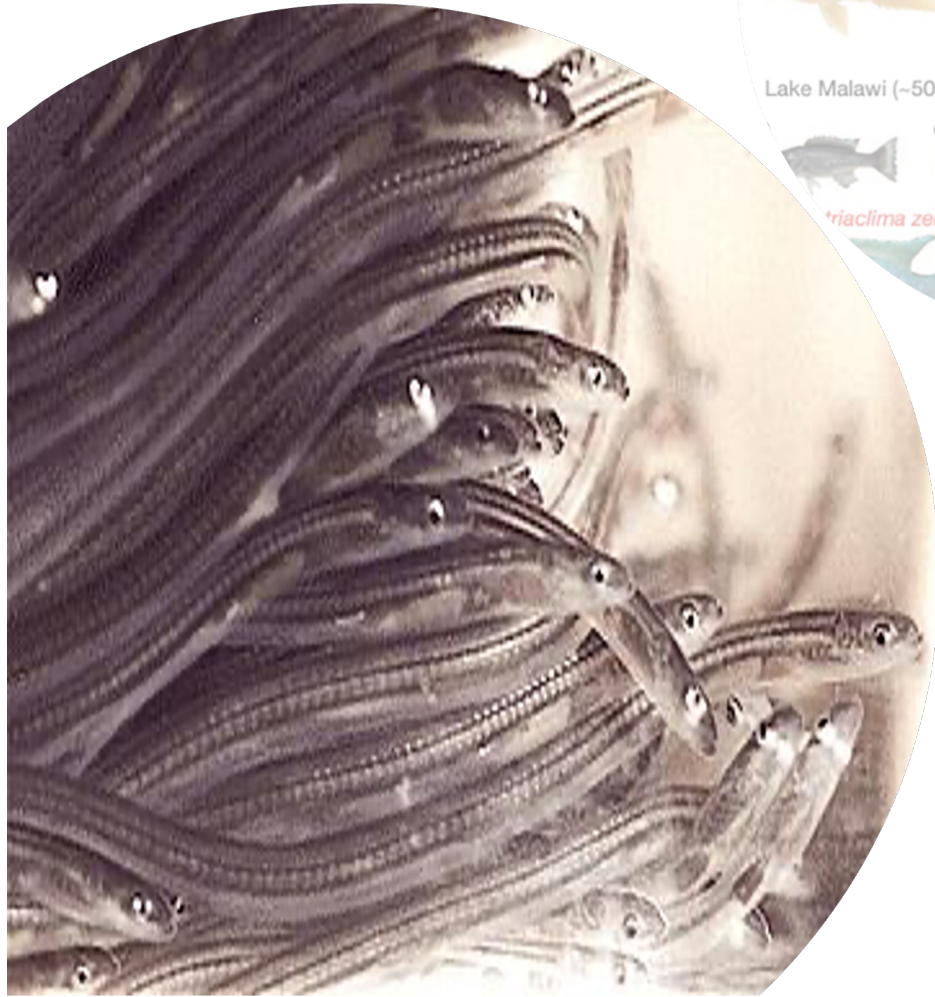
- cichlid_hierfstat.dat
Data for population analyses in fstat format.
- pop.txt
File with information about samples and populations.
You have to use this file to change the population column in the R object created when you read t
- Di_1-Di_2_global_mod.tsv
Pairwise FST from two species: limnothrissa black pelvic and macrops offshore. Are these two spec:

Correspondence between population number and Species names:

- 1 - macrops ngulube
- 2 - macrops black dorsal
- 3 - macrops offshore

Molecular Methods in Ecology and Evolution 2026





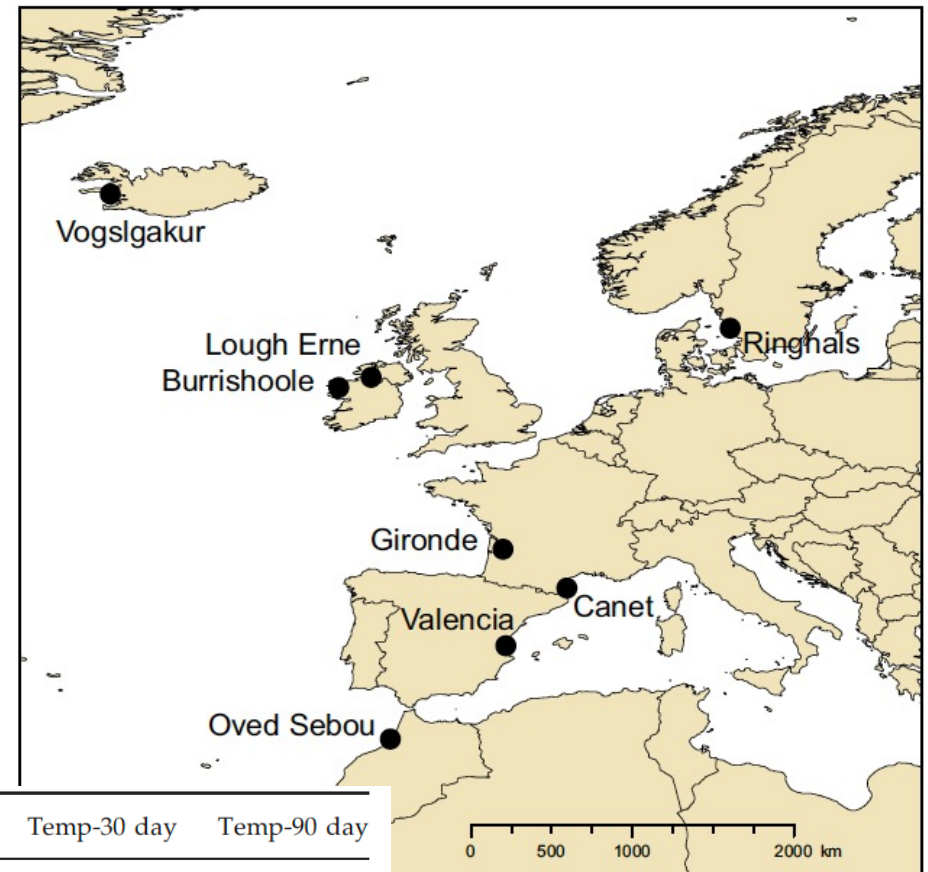
Project 4

The curious case of the European Eel.

Genome-wide single-generation signatures of local selection in the panmictic European eel

J. M. PUJOLAR,* M. W. JACOBSEN,* T. D. ALS,†‡ J. FRYDENBERG,* K. MUNCH,§ B. JÓNSSON,¶
J. B. JIAN,** L. CHENG,†† G. E. MAES,‡‡§§ L. BERNATCHEZ,¶¶ and M. M. HANSEN*

8 populations
240 individuals



Location	Code	N	Coordinates	Sampling date	Temp-10 day	Temp-30 day	Temp-90 day
Vogslækur, Iceland	ICE	34	64°69'N/22°33'W	2/7/2001	9.53	9.38	8.45
Ringhals, Sweden	RHG	30	57°21'N/12°27'E	15/3/2008	4.76	4.77	4.19
Lough Erne, Northern Ireland	LG	33	54°46'N/7°77'W	1/7/2008	13.93	13.81	13.85
Burrishoole, Ireland	BG	29	53°90'N/9°58'W	14/3/2005	9.36	9.79	9.57
Gironde, France	GG	37	44°86'N/0°42'W	16/4/2008	11.55	11.78	11.26
Canet, France	CAG	32	42°70'N/3°15'E	23/1/2008	12.73	12.61	13.24
Valencia, Spain	VG	31	39°46'N/0°24'W	15/1/2010	14.06	14.16	15.05
Oued Sebou, Morocco	MOR	33	34°26'N/6°70'W	28/4/2001	17.53	17.82	17.57

<https://github.com/Angelica-Pulido/MMEE-2026/>

Angelica-Pulido / MMEE-2026

Q

Type / to search

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Code

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Actions

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Wiki

Security and quality

Insights

Settings

Files

main

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Go to file

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1.Frogs_Sanger

>

2.Frogs_RADseq

✓

3.Cichlids

>

Data

3.0.README_Cichlids.md

✓

4.Eels

>

Data

4.0.README_Eels.md

4.0.README_Eels.md

.DS_Store

MMEE_2026_Manual.pdf

README.md

Schedule.png

MMEE-2026 / 4.Eels / 4.0.README_Eels.md

Angelica-Pulido

Update and rename 3.0.README_Eels.md to 4.0.README_Eels.md

197a5d9 · last year

Preview

Code

Blame

11 lines (7 loc) · 283 Bytes

Raw

Project: 4.Eels/

The files and folders for this project are as follow:

Data/

- eel_pops_lat_lon.tsv

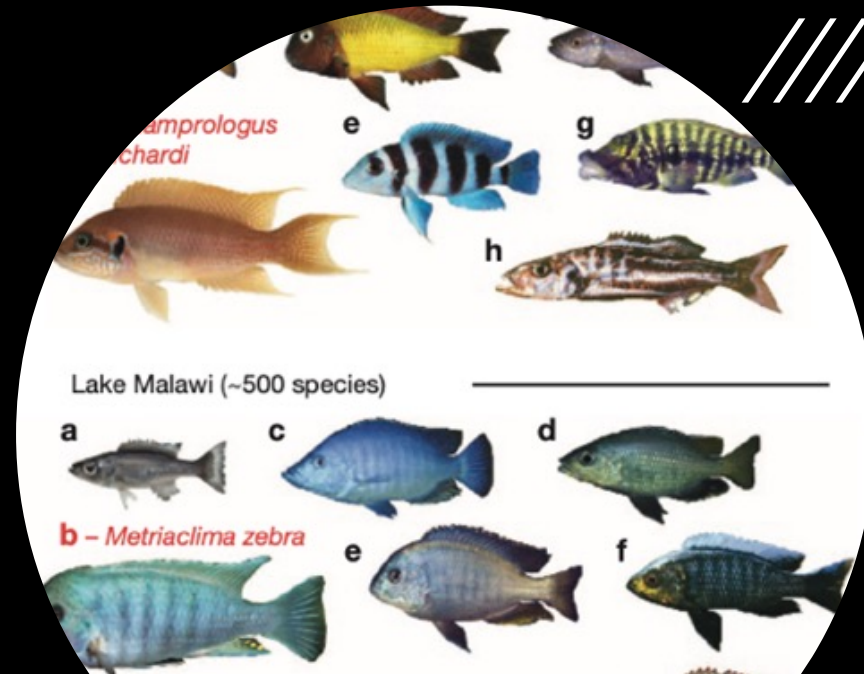
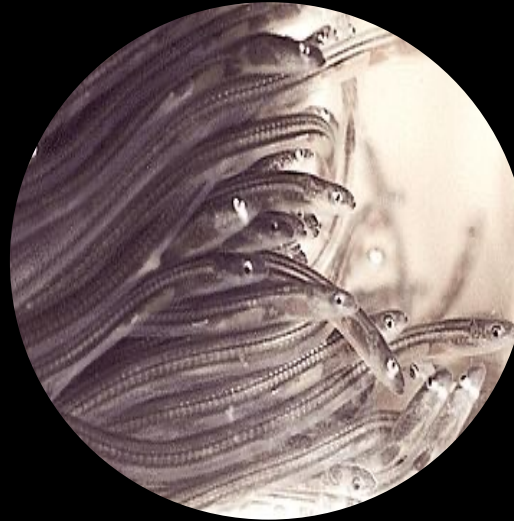
Coordinates of populations used in this project. You can use it to plot Eels populations on a map.

- eel_snps_subset.gen

genetic data for the eels in Genpop format.

Q.1. Can you explain the mysterious patterns of divergence?

Molecular Methods in Ecology and Evolution 2026





Project 1.

Use sanger sequencing on genetic markers to identify a mysterious frog species.

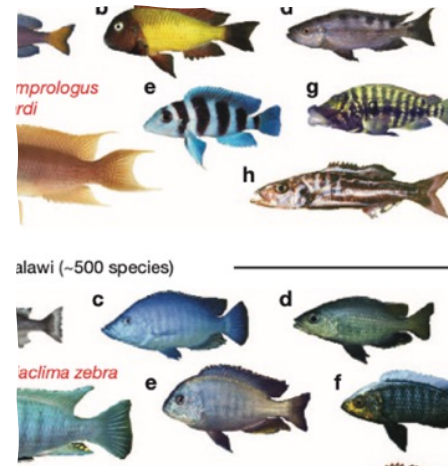


Project 2.

Cryptic speciation in tree frogs.

Allopatric divergence

Are they different species?



Project 3.

Divergence in Lake Malawi cichlids.

Ecological speciation in sympatry

What can be driving it?



Project 4.

The curious case of the European Eel.

Different types of **genetic markers**

- Single genes,
- RAD sequencing data

Different **methods**

- phylogeny
- Within- AND between-populations statistics
- clustering
- genome scans
- PCA
- Outlier Analyses ...

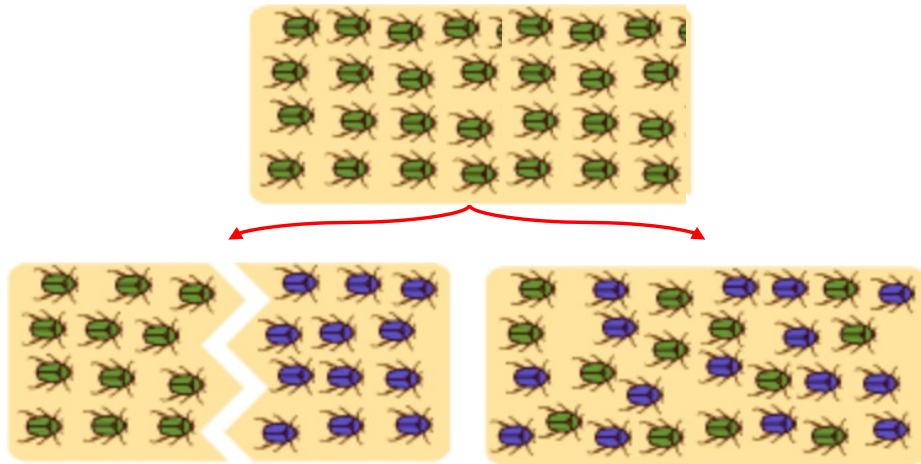
Different **geographic scales**

- Local (Fish)
- Regional (frogs)
- Continental (Eels)

The theme:

Understand the evolutionary and ecological drivers of population divergence and speciation

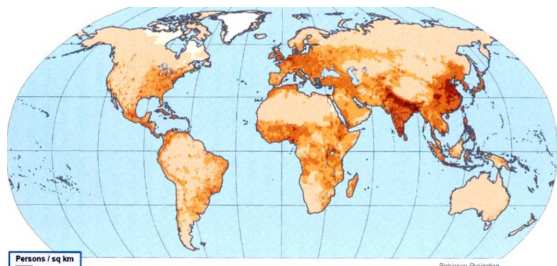
Initial population



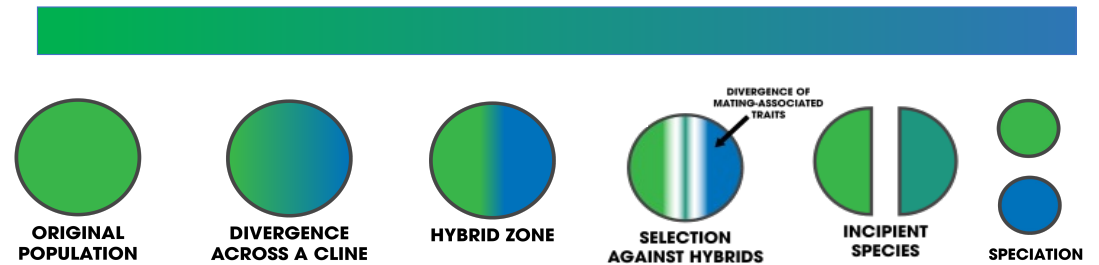
Geographical / Allopatric divergence

Ecological / sympatric divergence

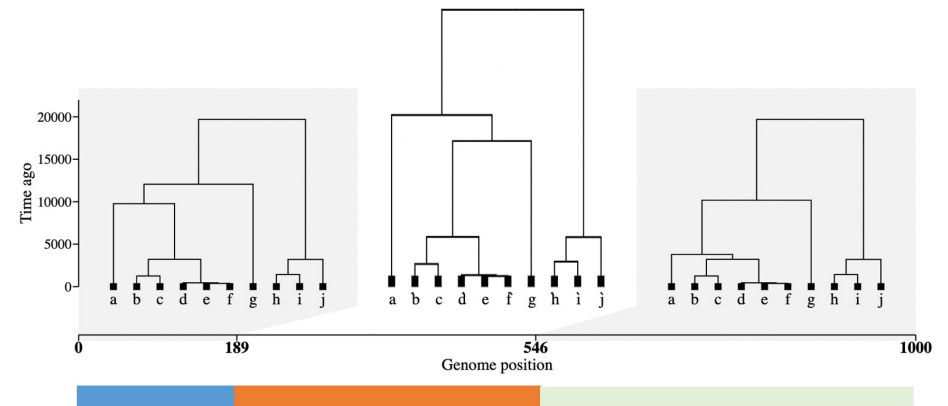
Eels?



The Speciation Continuum



Recombination



The report

General introduction
Wet lab – Computer Project 1 - Intro - methods - results & Discuss
Computer lab: Project 2. - Intro - methods - results - discussion - conclusion - references
Project 3. - Intro - methods - results - discussion - conclusion - references
Project 4. - Intro - methods - results - discussion - conclusion - references
General Discussion

max **5 pages** per project
(including text and figures)

max **25 pages** in total
(including text, figures and references)

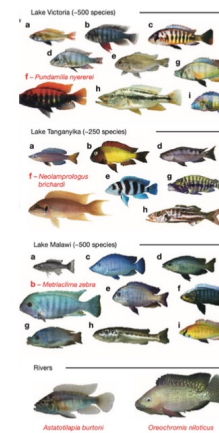
DEADLINE: 27 / 10 / 26

Report grading

See “**Advice on preparing your report**” section in your manual, these guidelines are there to help you write your report, but they will also be the basis of the marking scheme used to grade your reports.

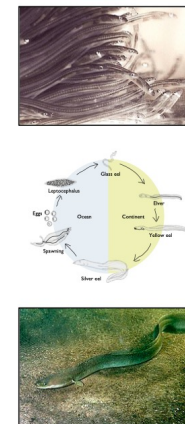
Using molecular approaches to understand the drivers of population divergence and speciation.

Cichlids

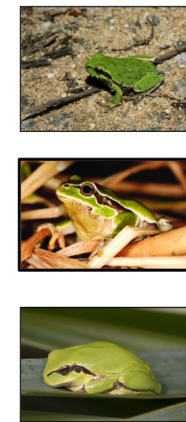


Brawand et al., 2014

European eel

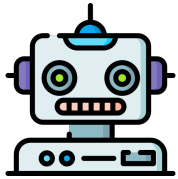


Hylid frogs



Prof. Ian R. Sanders, Dr. Luca Fumagalli, Prof. Nicolas Salamin

Teaching assistants: Dr. Angélica Pulido, Dr. Giulia Maiello, Marion Nyamari, Hugo Corval, Emelyne Gaudichau, Diego Clavijo.



Use of AI:

The goal:

You **learn** to analyze genomic data and interpreted results to answer biological questions.

DO NOT USE AI

- to write your report for you!
- to generate the interpretation or discussion of results.

YOU CAN USE AI

Clarify concepts that you don't understand

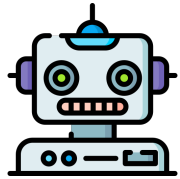
Organize workflow

Generate/**Interpret** bioinformatic code: Use it as a learning tool! – Copilot / Claude

Help in your code if you're stuck

Literature search

Use of AI:



Be critical!
Use it as a leaning tool!

AI tools are often incorrect!!

- Citation for specific claims
- Generating sentences that can not be verified by published studies
- Summarizing a scientific topic wrongly
- over-inflating importance of aspects
- constructing non-logical scientific arguments

Artificial intelligence at UNIL

Website: AI Insight

FAQ: Using AI at UNIL

- [FAQ: AI in Education](#)
- [FAQ: AI for my Studies](#)
- [FAQ: AI in Research](#)

AI_GO: Which AI tool for your data?

AI Training

AI Strategic Unit

AI Expert Group

FAQ: Using AI at UNIL

This FAQ is regularly updated (17/09/2026) to reflect the fast pace of developments in artificial intelligence. For any questions, contact the AI Strategy Unit: iaunil@unil.ch

[Table of contents](#) [Data protection and confidentiality](#)

Can AI be used at UNIL? ▾

What are the integrity values and recommendations for using generative AI at our university? ▾

ChatGPT, Large Language Models and AI: what are we talking about? ▾

What are the main language models currently available? ▾

Which AI tools does UNIL make available to its community? ▾

How can AI be used at UNIL? ▾

What are the risks of using AI tools and what precautions should be taken? ▾

What future directions can we anticipate in the development of this type of tool? ▾

<https://www.unil.ch/unil/en/home/menueinst/universite/intelligence-artificielle/faq.html>